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Review

Artificial intelligence and machine learning for production prediction and optimization of oil and gas surface networks

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HIGHLIGHTS

- AI/ML applications in oil and gas surface-network prediction are reviewed.
- The evolution from nodal analysis to physics-informed AI is traced.
- Major AI/ML families are compared by performance and deployment trade-offs.
- Digital twins are highlighted for real-time monitoring and decision support.
- Barriers in data quality, transferability, and trustworthiness are identified.

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ABSTRACT

Artificial intelligence (AI) and machine learning (ML) are transforming production prediction and optimization in oil and gas (O&G) surface networks, providing powerful alternatives to traditional physics-based models. This review provides a comprehensive overview of recent advancements in data-driven, hybrid, and physics-informed frameworks for surface network analysis, with a focus on their applications in forecasting, control, and decision-making. It traces the evolution of methods from early steady-state nodal analysis to digital twins and physics-informed AI, highlighting increased integration of real-time data, automation, and adaptive optimization. Key AI/ML techniques, including statistical learning, tree-based ensembles, deep learning, and neural operator architectures, are discussed in terms of predictive accuracy, scalability, and interpretability. The review also examines major challenges in implementation, including data quality, computational demands, model generalizability, and trustworthiness. Emerging areas, such as real-time adaptive control, hybrid AI-physics models, and closed-loop digital twins, are explored as tools for enabling autonomous and sustainable field operations. Overall, this review emphasizes that the future of surface network management will be driven by physics-informed, data-centric intelligence, supporting reliable, efficient, and understandable optimization across diverse oil and gas assets.

1. Introduction and background

The global energy landscape is currently undergoing a crucial transition, striking a balance between the urgent need for decarbonization and the persistent demand for reliable energy access. Despite the rapid rise of renewable energy sources, the oil and gas sector remains a vital part of the world's energy supply, providing essential fuels for transportation, power generation, and heavy industry [1]. In 2023, oil and natural gas accounted for approximately 53% of total primary energy demand, as depicted in Fig. 1, indicating the continued importance of hydrocarbons in maintaining global economic stability.

As industries worldwide face increasing pressure to boost efficiency and lower their carbon footprints, the oil and gas (O&G) sector has shifted its focus toward maximizing the recovery and efficiency of existing assets. In this context, optimizing surface production networks

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Nomenclature

AI	Artificial Intelligence
ANN	Artificial Neural Networks
API MPMS 10.10	API Manual of Petroleum Measurement Standards Chapter 10.10
API RP 90-1	American Petroleum Institute Recommended Practice 90-1
AR	Autoregressive
ARIMA	Autoregressive Integrated Moving Average
ARMA	Autoregressive Moving Average
BCP	Behind Casing Pressure
BHP	Bottom-Hole Pressure
BS&W	Basic Sediment and Water
BWRS	Benedict-Webb-Rubin-Starling
CatBoost	Categorical Boosting
CFR	Code Of Federal Regulations
CHP	Casing Head Pressure
CNN	Convolutional Neural Networks
DATS	Deep Autoregressive Time Series
DCA	Decline Curve Analysis
DRL	Deep Reinforcement Learning
DTR	Decision Tree Regression
EI HM 56	Energy Institute Hydrocarbon Management 56
FFT	Fast Fourier Transform
FNO	Fourier Neural Operator
GA	Genetic Algorithm
GLR	Gas-Liquid Ratio
GNN	Graph Neural Network
GPR	Gaussian Process Regression
GRU	Gated Recurrent Unit
HPC	High-Performance Computing
IEA	International Energy Agency
IPR	Inflow Performance Relationship
KNN	K-Nearest Neighbors
LGBM	Light Gradient Boosting Machine
LSB	Least Square Boosting
LSTM	Long Short-Term Memory
MA	Moving Average
MAASP	Maximum Allowable Annulus Surface Pressure
MAE	Mean Absolute Error
MAS	Multi-Agent Systems
MAWOP	Maximum Allowable Wellhead Operating Pressure
ML	Machine Learning
MLP	Multilayer Perceptron
MPC	Model Predictive Control
NMPC	Nonlinear Model Predictive Control
NPHI	Neutron Porosity
O&G	Oil And Gas
OCSVM	One-Class Support Vector Machine
ONN	Ordered Nearest Neighbor
PDE	Partial Differential Equation
PIML	Physics-Informed Machine Learning
PINN	Physics-Informed Neural Network
PINO	Physics-Informed Neural Operator
PSO	Particle Swarm Optimization
PVT	Pressure–Volume–Temperature
RF	Random Forest
RHOB	Bulk Density
RL	Reinforcement Learning
RMSE	Root Mean Square Error
RNN	Recurrent Neural Networks
RSM	Response Surface Methodology
SARIMA	Seasonal Autoregressive Integrated Moving Average
SCADA	Supervisory Control And Data Acquisition
SMAPE	Symmetric Mean Absolute Percentage Error
SVM	Support Vector Machine
SVR	Support Vector Regression
TAP	Tubing Annulus Pressure
TCN	Temporal Convolutional Network
THP	Tubing Head Pressure
THT	Tubing Head Temperature
TPDAT	Time Patch Dynamic Attention Transformer
VFM	Virtual Flow Metering
VLP	Vertical-Lift Performance
WNN	Weighted Nearest Neighbor
XAI	Explainable AI
XGBoost	Extreme Gradient Boosting

has evolved from a secondary operational task to a vital mechanism for ensuring safe, reliable, and economically sustainable operations. When executed effectively, optimization improves throughput, reduces downtime, and extends asset lifespan, benefits that are especially valuable for marginal or mature fields operating under tight economic constraints.

Surface production networks, especially in offshore settings, are among the most complex systems in modern engineering. These networks comprise interconnected wells, flowlines, risers, manifolds, separators, and compressors that must process multiphase fluids for export, as illustrated in Fig. 2. These systems necessitate substantial capital investment and exhibit strong interdependencies, rendering them highly

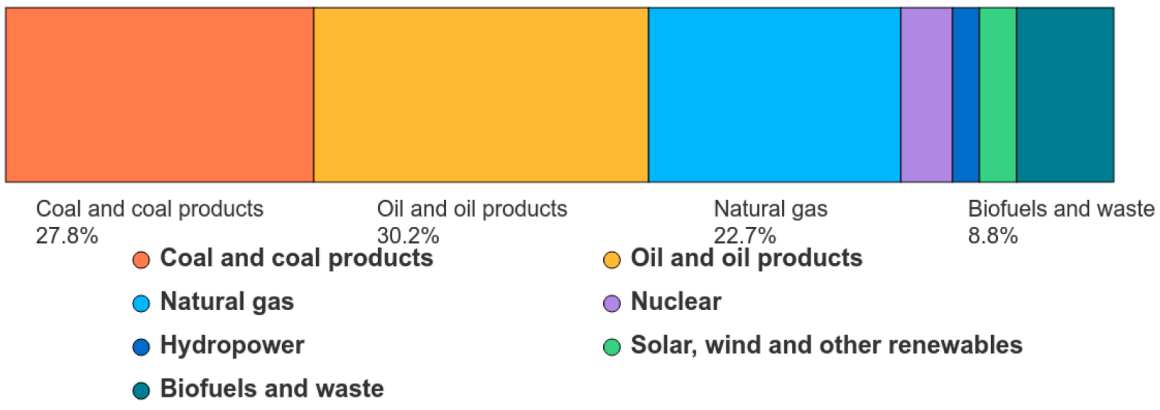


Fig. 1. Global primary energy mix in 2023, source: International Energy Agency [2].

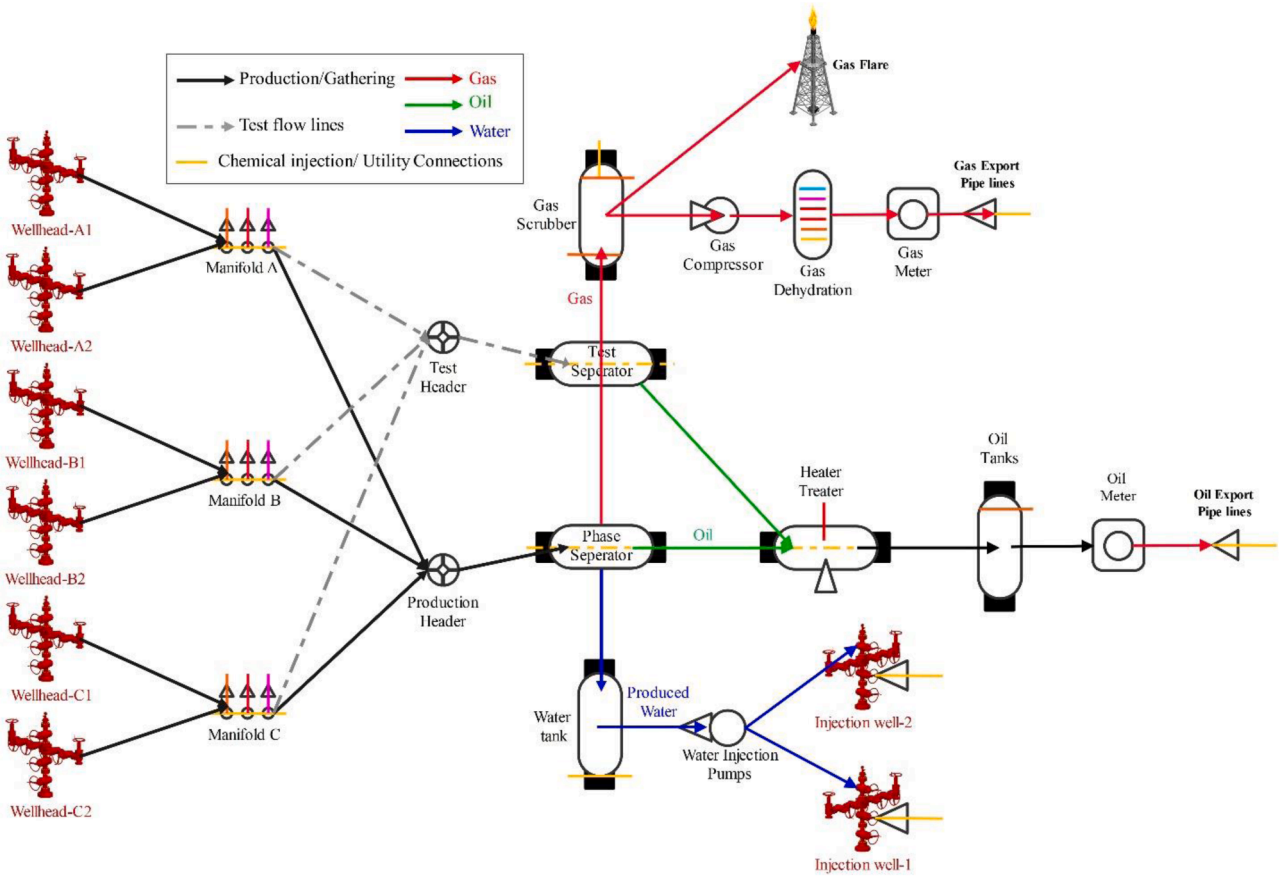


Fig. 2. Schematic representation of a typical surface production network showing the flow of oil, gas, and water through various processing and gathering components.

sensitive to operational decisions. Small actions, such as choke adjustments or pigging schedules, can spread through the field, impacting back-pressure, routing, and facility constraints in unexpected ways [3,4].

The physics of these networks is complex, with different flow regimes (stratified, slug, annular, and dispersed) emerging and disappearing as the rate, pressure, temperature, and pipe inclination change. Each regime has unique pressure-drop and transient characteristics that affect system stability [5]. Routine tasks, such as pigging, cause unsteady surges and thermal or hydraulic shifts that stress the infrastructure [6]. Operators must manage threats like wax and asphaltene deposition, hydrate formation, and sand production. To address these, they use chemical, solvent, and thermal treatments, such as hot-oil or hot-diesel circulation, to prevent deposition or viscosity-related losses [7,8]. If these threats are not anticipated and managed, they can lead to costly shutdowns and accelerate equipment degradation [9]. In this environment, optimizing surface networks is crucial for ensuring safe, reliable, and economically robust operations. As was previously highlighted, effective optimization boosts throughput, minimizes downtime, and prolongs asset life, which are especially vital benefits for mature fields [10].

The conceptual development of surface-network engineering reflects the industry's operational needs. Early nodal analysis relied on empirical correlations and drift-flux concepts. Mechanistic two-fluid models restored physical consistency across different flow regimes. Steady-state network solvers combined wells, flowlines, and facilities into comprehensive tools for field-scale deliverability. As deep-water developments and long tiebacks expanded, high-fidelity transient simulators emerged to accurately model terrain-induced slugging, cooldown and restart behavior, and wax or hydrate risks. Validated software, such as OLGA [11] (for dynamic multiphase simulation) and PIPESIM [12] (for network analysis and planning), has become an industry standard. The main value now lies in making minute-to-hour decisions under uncertainty and constraints. However, traditional tools consistently face three limitations: calibration fragility (field-specific closures for interfacial friction, heat transfer, and deposition kinetics often lack instrumentation), computational latency (dynamic analysis can be too slow for operational cycles), and model-plant mismatch (worsened by changing fluids, aging assets, and sensor drift). As assets become more interconnected, these constraints lead to conservative set points and slow responses to disturbances, resulting in missed economic opportunities.

Concurrently, O&G operations generate a vast amount of high-frequency measurements, including SCADA (supervisory control and data acquisition) streams, historian archives, well-test and PVT (pressure-volume-temperature) data, maintenance logs, pigging and chemical schedules, and metocean records. This situation presents a significant opportunity to integrate data-driven intelligence with first-principles simulation.

Artificial Intelligence (AI) and Machine Learning (ML) can learn nonlinear response surfaces, infer unmeasured states, and suggest actions in the face of uncertainty, thereby complementing and improving physics-based models [13]. This includes ensembles aimed at improving prediction reliability; sequence models like Long Short-Term Memory (LSTM), Temporal Convolutional Network (TCN), and Transformer for transient forecasting, Graph Neural Networks (GNNs) that follow network topology, physics-informed surrogates such as Physics-Informed Neural Networks (PINNs), and neural operators like Fourier Neural Operator (FNO) and Physics-Informed Neural Operator (PINO). These methods significantly cut simulator runtimes while ensuring compliance with governing equations. Related work [14] in

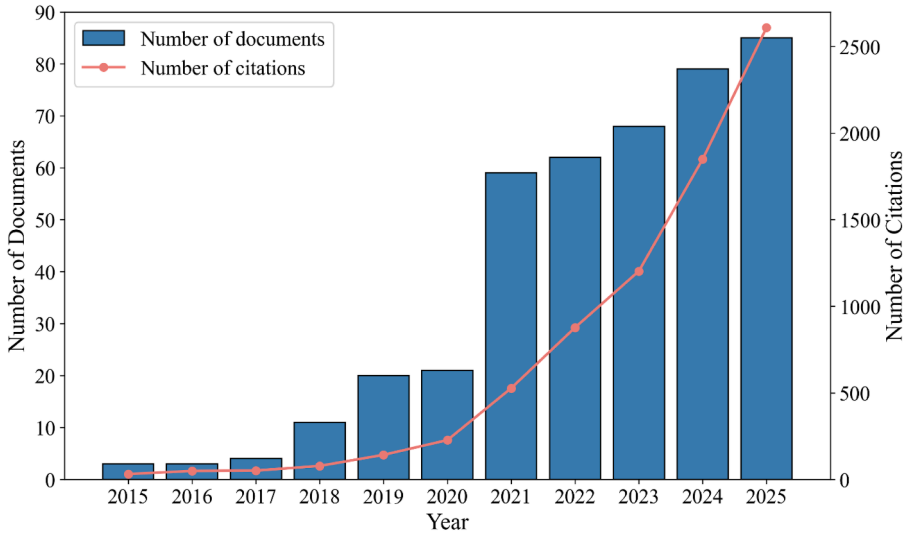


Fig. 3. Annual trends in publications and citations related to AI and ML applications in the oil and gas sector from 2015 to 2025, based on data retrieved from the Web of Science.

unconventional reservoirs has also shown that combining geological descriptors with operational variables can improve shale-gas production forecasting, highlighting the importance of integrating both subsurface and production-history information in data-driven productivity prediction. Collectively, they enable AI-enhanced digital twins for virtual flow metering (VFM) in situations where physical meters are impractical, supporting short-term transient forecasting, nonlinear model-predictive control, and real-time network optimization. In this context, a digital twin is a continuously updated, data-driven representation of the physical production system that combines measurements and physics-based models to reflect current surface network behavior and support day-to-day operational decisions.

Fig. 3 illustrates the yearly number of publications and citations related to artificial intelligence and machine learning applications in oil and gas production from 2015 to 2025, as sourced from the Web of Science. The figure clearly shows a strong upward trend in both research output and scholarly impact over the last decade. Publication activity was modest before 2020 but started to increase sharply afterward, signaling growing global interest in AI/ML integration in the oil and gas industry.

Despite the rapid growth of AI/ML research in oil and gas, both the literature and field adoption remain fragmented. Published studies are dispersed across model families, operational tasks, and asset scales, while deployed solutions are often isolated point applications rather than integrated decision-support systems across wells, manifolds, and facilities. In many cases, studies are presented either as standalone forecasting or optimization applications, or as algorithm-centered contributions without a unified view of prediction, assurance, and control across interconnected surface facilities. Moreover, uncertainty quantification and governance are frequently underspecified, and credible evidence for safe closed-loop use beyond pilot studies remains limited. Together, these gaps hamper trust, portability across assets, and the measurement of operational, economic, and emissions-related impact, creating a clear gap between methodological progress and the practical needs of surface-network management. Existing reviews on AI and ML in upstream oil and gas have also often focused on a single methodological family, a single asset scale, or a single application task. In contrast, the present review takes an end-to-end surface-network perspective, covering the chain from wellhead measurements and soft sensing to network-level monitoring, forecasting, optimization, and decision support.

To address this gap, the present review is guided by the following objectives and questions:

1. To identify and organize the principal AI/ML model families applied to oil and gas surface-network problems, including production prediction, assurance, and control/optimization.
2. To compare these approaches in terms of reported performance, data requirements, interpretability, physical consistency, and deployment relevance.
3. To define the scope boundaries of the review by focusing on oil and gas surface production systems, while excluding studies that deal exclusively with subsurface reservoir simulation without a direct surface-network connection.
4. To identify the most important technical and practical research gaps that currently limit trustworthy, field-scale adoption, including issues of data quality, transferability, uncertainty awareness, and operational integration.

This structure supports a critical, task-oriented synthesis of the literature and helps clarify both benchmarking needs and deployment relevance.

Consistent with these objectives, this review critically examines AI/ML applications for prediction, assurance, and control in surface production networks, tracing the evolution from empirical correlations and mechanistic models to AI-enhanced digital twins and physics-informed machine learning. The review covers statistical learning, deep sequence models, graph neural networks, hybrid gray-box approaches, and physics-guided surrogates for virtual flow metering, short-term forecasting, anomaly detection, and nonlinear model predictive control (NMPC). It also considers practical deployment issues, including edge, on-premises, and cloud implementation, together with drift monitoring, auditable data lineage, and protected actuation.

This review paper categorizes the literature into three main areas: production prediction, assurance (including anomaly detection, fault detection, and soft sensing), and control/optimization. It links each task to suitable model types such as ensembles, sequence models, graph neural networks, and hybrid or physics-informed approaches. Reported performance is synthesized using standard evaluation metrics together with deployment-relevant considerations, highlighting where data-driven methods outperform or complement traditional mechanistic models. The review also discusses integrity-aware control strategies,

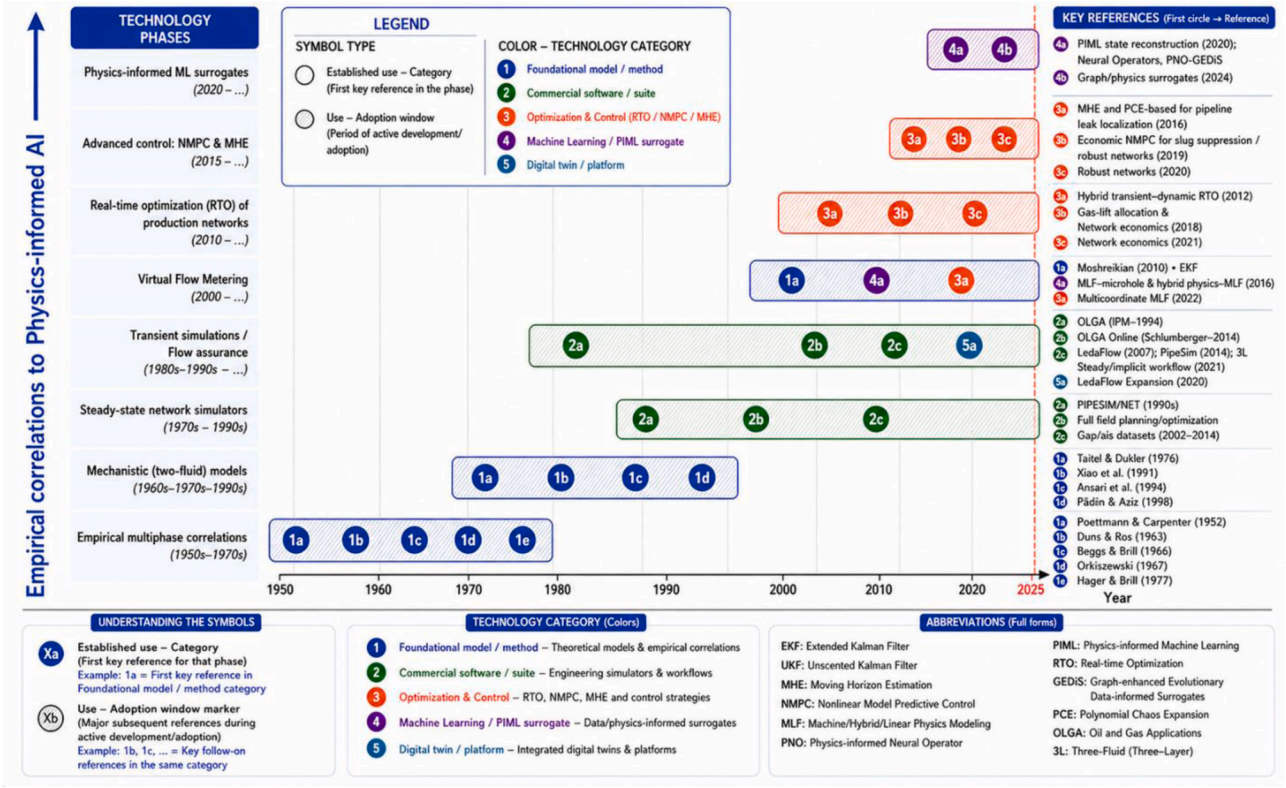


Fig. 4. Methodological evolution of surface-network prediction and optimization, showing the progression from empirical and mechanistic modeling to steady-state and transient simulation, virtual flow metering, real-time optimization, advanced control, and physics-informed AI approaches.

calibrated uncertainty quantification, human-in-the-loop safeguards, and post-action auditing needed for defensible closed-loop operation. Overall, it emphasizes evidence, safety, and operational practicality to clarify where AI provides lasting value in surface-network management.

The primary AI/ML review corpus used in this paper was identified through structured searches of Web of Science Core Collection, Scopus, OnePetro, IEEE Xplore, and ScienceDirect, covering publications from January 2015 to March 2026. The search strategy combined oil and gas surface-network terms, including "oil and gas", "surface network," "production network," "virtual flow metering," and "digital twin," with artificial-intelligence terms such as "machine learning," "deep learning," "neural network," "physics-informed," "neural operator," "graph neural network," "transformer," and "reinforcement learning." Retrieved records were screened for relevance to oil and gas surface production prediction, monitoring, assurance, control, or optimization. Studies were retained when they reported methodological development, comparative benchmarking, or field-oriented application and provided sufficient technical detail. Studies focused exclusively on subsurface reservoir simulation without a clear link to surface-network decisions, as well as editorials, duplicate records, and incomplete preprints, were excluded. Earlier seminal references on multiphase-flow modeling, production systems, and control were additionally retained where necessary to provide historical and methodological context for the evolution discussed in Section 2. The final body of literature was then categorized by application task and model family, forming the basis for the synthesis presented in Sections 3.2–3.6 and the comparative summary table. In methodological terms, the present manuscript adopts a hybrid review approach that combines structured database-based literature screening with thematic content synthesis, following recent review practices that use thematic grouping to strengthen analytical rigor and identify research trends and gaps more explicitly [15–17].

2. Evolution of modeling and simulation methods for the oil and gas industry

Surface-network flow modeling in the oil and gas industry has shifted from empirical correlations based on laboratory data to the use of physics-informed, learning-based surrogates that operate in real time, as shown in Fig. 4. Initial methodologies depended on vertical and inclined-pipe correlations, including those by Duns & Ros [18], Hagedorn–Brown [19], Orkiszewski [20], and Beggs–Brill [21], frequently justified using the drift-flux concept proposed by Zuber & Findlay [22]. The methods demonstrated speed and were suitable for nodal analysis; however, they showed fragility outside their calibration ranges. From the late 1970s through the 1990s, mechanistic models (two-phase models) of fluid systems with explicit regime diagrams incorporated force balance considerations (e.g., Xiao–Shoham–Brill [23], Ansari et al. [24], and Petalas–Aziz [25]). They enhanced the capacity for extrapolation across variations in diameter, inclination, and fluid characteristics. Their contributions supported steady-state network solvers and, with the advent of dynamic two-fluid simulators, particularly OLGA, opened the flow-assurance era, enabling analysis of severe slugging, cool-down/restart, and hydrate/wax risks [11]. With the proliferation of instrumentation, virtual flow metering integrated physical principles with estimation and data-driven components for continuous allocation. Subsequently, models were incorporated into operational procedures through real-time optimization and model-predictive control. The current frontier involves physics-informed machine learning, including PINNs for state reconstruction, neural operators for solution maps, and physics-informed graph neural networks for topology-aware production forecasting. These approaches promise significant speed improvements while maintaining physical constraints [26–28]. Table 1 critically assesses the methodological evolution of oil and gas surface-network models, their adaptation, and execution. As summarized in Fig. 4, the

Table 1
Methodological evolution of oil-and-gas surface-network models and their adaptation and execution.

Method	Application	Accuracy in Physics	Execution	Data requirements	Advantages	Failure Modes and Caveats
Empirical multiphase correlations (A/ B/C)	Preliminary nodal analysis	Low to moderate	Fast	Minimal	Basic and readily accessible	Error due to out-of-range values, transitions between regimes, restricted applicability [21]
Mechanistic (two-fluid) models	Comprehensive design; broad diameter/ inclination	High	Medium	Moderate	Improved extrapolation, physical clarity	Closure sensitivity is associated with increased computational demands and complexity [23]
Steady-state network simulations	Comprehensive planning across the field, identification of bottlenecks, and static allocation of gas-lift resources	High (Steady)	Medium	Moderate	Constraints at the network level and extensive topologies	Absence of dynamics, initiation or restart requires transient tools
Transient simulators/ flow assurance	Flow assurance in pipelines to prevent slugging during system restart operations	Highest (1-D two-fluid)	Slow to Medium	Moderate	Standard practices for essential operations in the industry	The challenges include high computational demand, parameter identification, and discrepancies between the model and platform data [11]
Virtual Flow Metering	Soft sensing of rates employing model-based approaches or ML	Model-based high performance, ML outcomes vary significantly	Real-time	Sensors and historical data	Comparison of low-cost options and MPFMs, analysis of continuous rates	Quality of sensors, robustness across various regimes, tuning mechanisms [29]
Real-time Optimization (RTO) of production networks	Optimization of economic set-points, including gas-lift allocation, choke, compressor, separator targets, and routing	Moderate to high (steady-state or hybrid)	Online	Real-time SCADA/ historical data, operational limits, PVT analysis	Coordination across the field, clear economic principles, manages discrete decision-making	Mismatch in time constants, outdated economic models, unaccounted constraints, local optima [30]
Advanced control: NMPC/ MHE	Closed-loop multivariable control and estimation, slug suppression, gas-lift stabilization	High (dynamic)	High	High-rate sensors, actuator models, constraints	Impose constraints, model transients, incorporate soft sensing, and enhance throughput and energy efficiency	Model discrepancies, estimator drift, solver infeasibility, and missed deadlines necessitate fallback mechanisms [31]
Physics-informed ML surrogates	Quick scenario-analysis, internal iterations for RTO/ MPC, inverse problem solving	High, provided it is well-constrained	Fast after training	High (simulation or field)	Significant speed improvements	Objective-function/design-space fragility, implementation of rigid physical principles, uncertainty quantification is not straightforward [26,28].

Legend: Category A = Homogeneous models, which do not consider phase slip, Category B = Model that represents slip but do not consider flow regimes (phase distributions in multiphase flow), Category C = models that incorporate phase slip as well as flow regimes, MPFMs = Multiphase flow meters, NMPC = Nonlinear model predictive control, MHE = Moving horizon estimation.

key transition is not merely chronological but functional. The field has moved from offline design and correlation-based prediction toward faster, adaptive, and increasingly physics-constrained AI frameworks intended to support operational decision-making under uncertainty.

3. AI/ML Techniques for surface networks

The integration of AI/ML into oil and gas surface network optimization represents a paradigm shift from conventional physics-based modeling to advanced data-driven approaches. Fig. 5 depicts the schematic of the data-driven modeling workflow for production forecasting and optimization. The framework includes data preprocessing, feature engineering, model training and validation, followed by real-time deployment and decision support. A feedback loop enables continuous model updating as new operational data become available. Unlike mechanistic simulators, which rely on deterministic, first-principles formulations [29], AI/ML models utilize large-scale datasets from sensors, SCADA systems, and historical operational records to learn complex, nonlinear system dynamics. These methods can capture complex nonlinear dynamics, adapt to changing field conditions, and deliver near-real-time predictions to optimize production systems. In surface network applications, AI/ML models have demonstrated their ability to enhance production rates, reduce lifting costs, improve equipment reliability, and minimize downtime through anomaly detection, trend forecasting, and predictive maintenance. This section discusses AI/ML techniques by task, including virtual flow metering (VFM), forecasting, anomaly detection, and network-level optimization and control.

3.1. Oil and gas production data

Monitoring of production wells is predominantly conducted via measurements at the wellhead and near-wellhead locations. Wellhead measurements and choke control in oil and gas (O&G) production systems, as illustrated in Fig. 6, are crucial for determining the system state, establishing the feasible control envelope, and identifying optimization levers. The variable under manipulation is the choke size (“bean” 1/64-in units), which establishes a controllable nonlinear pressure drop at the wellhead to manage drawdown and mitigate riser-induced slugging [32]. Critical parameters include tubing head pressure (THP), which connects reservoir inflow and vertical lift to surface network outflow, and tubing head temperature (THT), which is the most immediate indicator of hydrate and wax risk [33]. Annulus pressures, including tubing annulus pressure (TAP), casing head pressure (CHP), and behind-casing pressure (BCP), serve as integrity constraints. Additionally, basic sediment and water percentage (BS&W) quantifies the produced sand and water fraction, which influences separator loading [34]. Table 2 provides a comprehensive summary of the physical significance of each variable and its corresponding control function, along with relevant regulatory and metrology references for annulus management (API RP 90–1, 30 CFR 250) and BS&W (API MPMS 10.10 / EI HM 56).

In oil and gas surface production, fluid flow through tubing and flowlines is governed by frictional, gravitational, and acceleration effects, which depend on gas-liquid ratio (GLR), flow rate, pipe inclination, mixture density/holdup, and gas-liquid slip. The production system’s operating point is defined by the intersection of the inflow

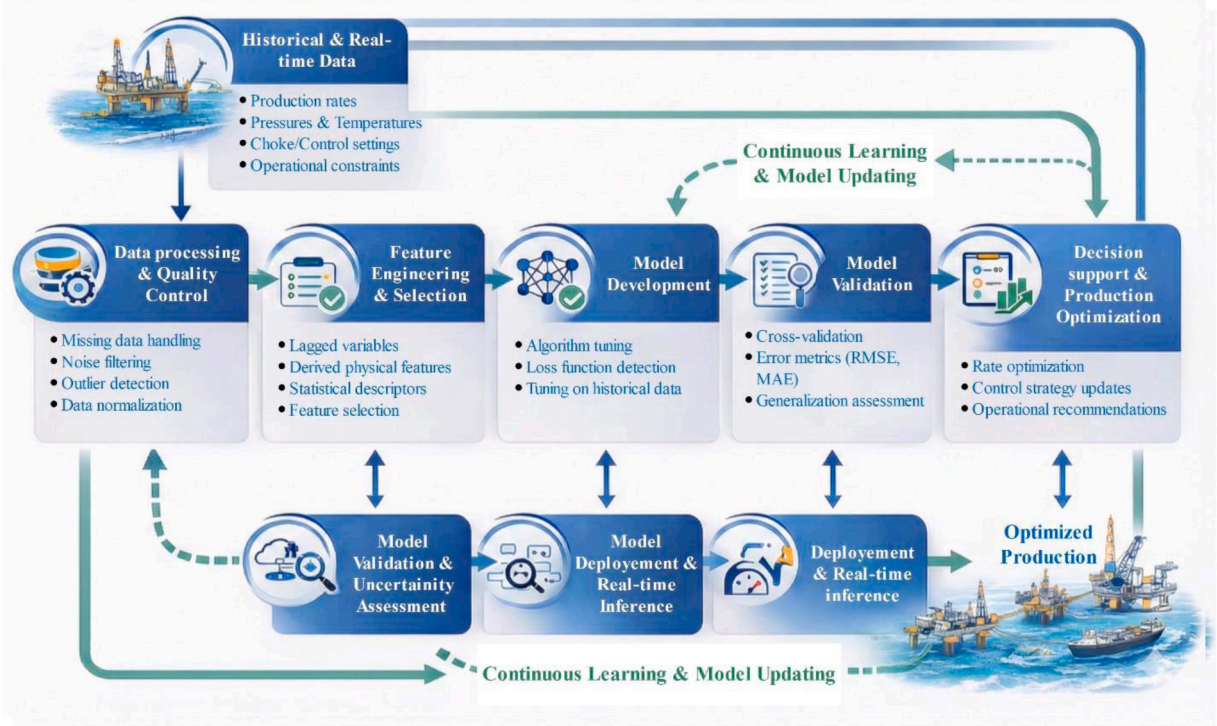


Fig. 5. Data-driven modeling framework for production optimization, illustrating the workflow from data acquisition and preprocessing to model training, validation, deployment, and decision support. The feedback loop emphasizes that AI/ML models in surface networks must be continuously updated as new operational data become available.

performance relation (IPR) and vertical lift performance (VLP) curves. THP and choke size shape the VLP, while reservoir pressure and relative permeability determine the IPR. The choke is the main control element. Opening it increases liquid and gas flow, lowers THP, increases drawdown, and often cools the system (e.g., via Joule–Thomson effects). Closing it has the opposite effect: higher THP, reduced drawdown, lower rates, and reduced slugging [33]. Because choke hydraulics are nonlinear, even small changes can have a significant impact. In a gas-dominated flow, if the choke's downstream-to-upstream pressure ratio falls below a critical value, the flow becomes choked and independent of downstream pressure [32]. Since no universal correlation works under all conditions, operators regularly update choke maps and account for uncertainty. Annulus pressures change with temperature due to thermal breathing, and BS&W can increase if high drawdown induces water coning. Facility constraints must be respected, such as separator limits, residence time, annulus pressure thresholds, and casing pressure requirements. In multi-well systems, coordinated choke adjustments can reduce slugging but may increase manifold pressure. Ultimately, optimal production depends on balancing choke size, pressure, and temperature to maximize oil/gas production rates while minimizing equipment stress, flow instabilities, and operational risk. This approach ensures that physics-based control from THP, THT, TAP, CHP, BCP, and BS&W is in accordance with operational compliance, as detailed in Table 2.

3.2. Statistical machine learning methods

Statistical modeling captures system behavior through mathematical relationships among variables, providing a foundation for both regression-based inference and time-dependent forecasting. While classical regression assumes independence among observations, time-series methods explicitly incorporate temporal dependencies to enhance predictive accuracy and reveal underlying system dynamics. In oil and gas production analytics, widely applied temporal models include autoregressive (AR), moving average (MA), autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), and their seasonal variants (SARIMA). Considering the use of statistical ML methods in oil and gas production prediction, Qiao et al. [35] proposed a hybrid forecasting model that integrated Particle Swarm Optimization (PSO) with Least Squares Support Vector Machine (LS-SVM) to improve prediction accuracy. Using historical production data from a Chinese oil and gas company, they built a data-driven model that preprocesses the data to remove anomalies and automatically optimizes model parameters. Simulations conducted in MATLAB demonstrated that the PSO-LS-SVM model produced forecasts closely matching actual production data, showing high convergence speed and prediction precision, and demonstrating its suitability for real-time production forecasting and optimization tasks. Similarly, Panja et al. [36] developed and compared three AI surrogate models: Response Surface Model (RSM), LS-SVM, and Artificial Neural Networks (ANN), to forecast hydrocarbon production from hydraulically fractured shale reservoirs. They utilized a synthetic dataset derived from 144 commercial black-oil simulations (IMEX) and included eight key input parameters: reservoir permeability, initial gas–oil ratio, rock compressibility, gas relative permeability exponent, slope of the gas–oil ratio, initial pressure, bottom-hole pressure, and fracture spacing. Results indicated that both RSM and LS-SVM achieved strong oil recovery predictions, with LS-SVM outperforming the others in modeling the complex behavior of gas–oil ratios. ANN performed adequately but was less robust for extrapolation. The study's main limitation lies in its reliance on simulated rather than field data and its neglect of physical flow mechanisms, which may reduce model generalizability.

Tree-based ML methods such as Decision Tree (DT) and Random Forest (RF) algorithms are highly valued for their interpretability, capacity to handle mixed data types, and robustness in capturing nonlinear relationships, rendering them powerful tools for predictive modeling in drilling operations, production forecasting, reservoir characterization, and pipeline integrity management. Combining tree-based models with

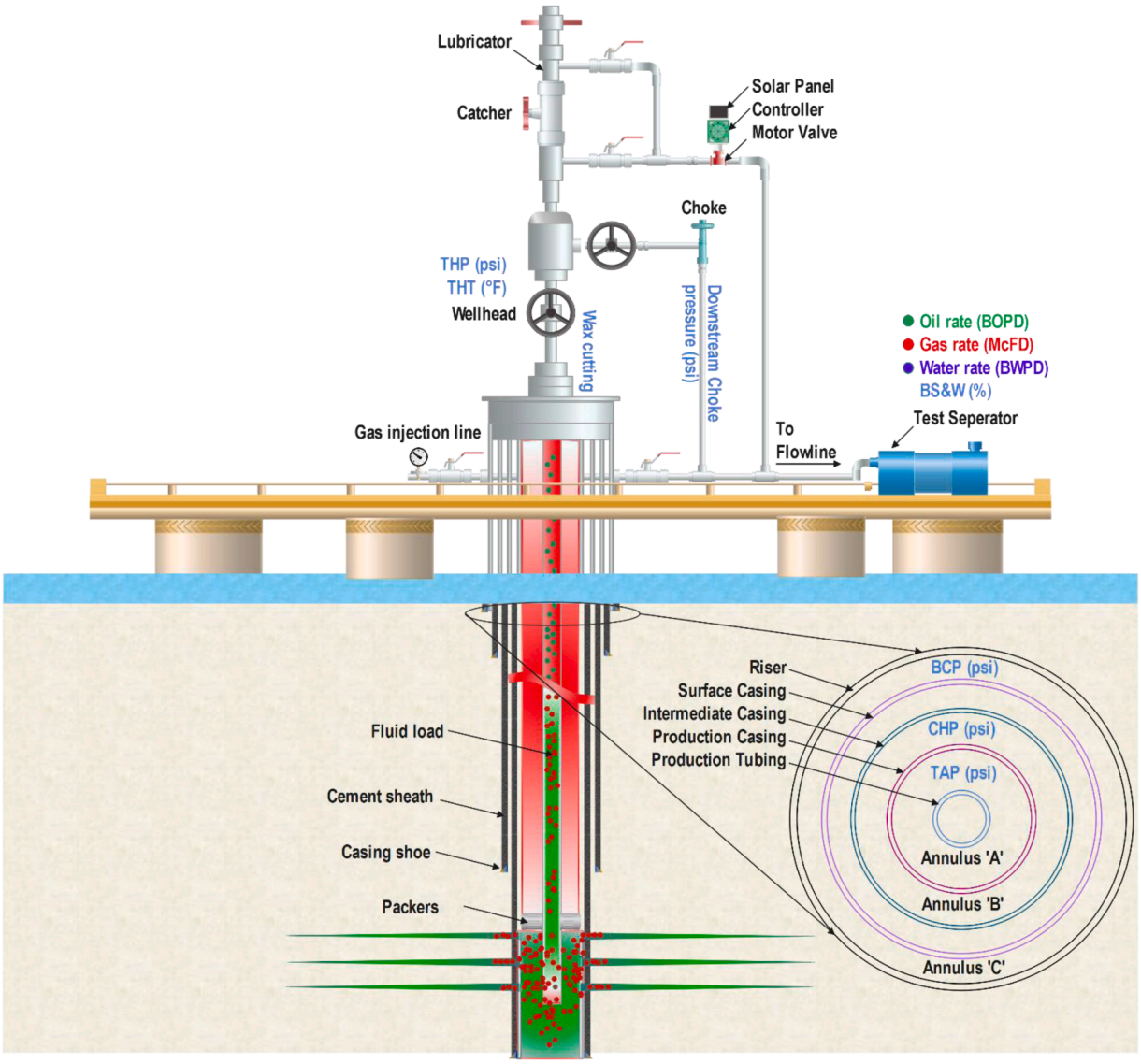


Fig. 6. Schematic representation of data measurement points in a surface production network, as well as typical sites for maintenance operations.

other ensemble or optimization methods has made oil and gas production systems more reliable and accurate in predictions. In the context of pipeline safety and reliability, Mazumder et al. [37] developed a non-temporal classification model using multiple ML algorithms: K-Nearest Neighbors (KNN), DT, RF, Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LGBM), and Categorical Boosting (CatBoost), to assess pipeline failure risks. Their dataset comprised 959 samples containing parameters such as pipeline diameter, wall thickness, defect geometry, tensile strength, and operating pressure. Among the models evaluated through accuracy, precision, and recall metrics, XGBoost achieved the best performance with an 85% accuracy, demonstrating the utility of ensemble tree-based techniques for pipeline integrity assessment. Liu et al. [38] investigated pipeline defect prediction using 1500 well log samples from North China, evaluating models including Logistic Regression (LR), SVM, Gaussian Process Regression (GPR), and DT-based ensembles. The ANN model achieved the highest accuracy (coefficient of determination R^2 = 99% for training, 96% for testing), but RF and DT approaches provided interpretable and efficient alternatives for non-temporal defect assessment. Similarly, Ma et al. [7] applied RF, XGBoost, SVM, and LGBM to estimate burst pressure in corroded oil and gas pipelines, achieving an impressive R^2 of 99% in training and 98% in testing using 314 samples characterized by pipeline geometry, material strength, and corrosion depth.

RF has also been applied in reservoir simulation and production prediction. Kumar and Hassanzadeh [39] utilized RF to analyze 2D reservoir simulations (240 temporal samples), predicting the performance of shale barriers based on physical and thermal properties (e.g., formation compressibility and thermal conductivity). The model achieved high R^2 values and low root mean square error (RMSE), underscoring RF's adaptability to dynamic reservoir systems. Similarly, Otchere et al. [40] used Bayesian-Optimized XGBoost (BayesOpt-XGBoost) on Equinor Volve Field data to predict porosity, shale volume, and water saturation. Their hybrid model achieved an overall accuracy of 93% and a precision of 98%, validating its potential for reservoir characterization and production forecasting.

In well integrity and production fault detection, Santos et al. [41] used the publicly available Petrobras 3W dataset [42], which contains labeled multivariate time-series data. They employed random forests to classify undesirable operating events using variables such as gas-lift choke pressure, downstream temperature, and flow rate. The model achieved 94% accuracy, demonstrating its ability to identify early production anomalies and enhance operational reliability. In an investigation of pipeline corrosion rate forecasting, RF and CatBoost were applied

Table 2
Surface network variables, physical meaning, and control function.

Variable	Physical location/ interpretation	Key functions & control
Choke size (1/64 of an inch)	Effective flow area at the surface choke	Regulates wellhead pressure drop, controls production rate, stabilizes flow, and protects facilities.
Tubing head pressure (THP)	Upstream pressure at the wellhead/tubing	Critical coupling variable between subsurface inflow and surface network hydraulics, safeguards separators and headers within facility constraints.
Tubing head temperature (THT)	Fluid temperature upstream of the choke	Indicates hydrate and wax risk, influences viscosity and phase behavior, guides chemical and thermal interventions.
Downstream choke pressure (DSCP)	Flowline or separator inlet pressure	Determines flow regime (critical/sub-critical), mediates well interference, and constrains separation efficiency.
Tubing annulus pressure (TAP)	Pressure in tubing-casing annulus (Annulus 'A'*)	Monitors integrity for thermal effects, migration, or leaks, regulated by API RP 90-1 and 30 CFR 250.
Casing head pressure (CHP)	Pressure at casing head (Annulus 'B'*)	Indicates barrier health and communication status, constrained by MAWOP/MAASP and diagnostic thresholds.
Behind-casing pressure (BCP)	Pressure trapped behind casing strings (Annulus 'C'*)	Categorizes sustained, thermal, and operator-imposed pressures, guides remediation and operating limits per RP 90–1.
Basic sediment & water (BS&W)	Sand and water fraction in produced liquids (online analyzer or lab)	Determines separator and pump loading, ensures compliance with produced water standards (API MPMS 10.10 / EI HM 56).

* Annuli A, B, and C are highlighted later in Fig. 6.

to 3240 non-temporal samples with predictors such as gas composition, pressure, and velocity [43]. CatBoost achieved the highest accuracy (R^2 = 99.9%), demonstrating the efficiency of gradient-boosted tree ensembles for estimating corrosion tendencies in O&G systems. In another non-temporal pipeline failure analysis, a large dataset of 149,940 records was processed using Multilayer Perceptron (MLP), RF, and SVR models [44]. The features included operational, safety, and maintenance indicators reflecting transportation disruptions and equipment reliability. The models were benchmarked for computational efficiency and prediction accuracy, with the MLP and SVR achieving competitive performance while reducing computation time, underscoring their applicability to large-scale infrastructure monitoring. Corrosion modeling within oil pipeline systems has also benefited from hybrid AI approaches. Naserzadeh and Nohegar [45] applied a suite of metaheuristic-enhanced SVR models, optimized using Genetic Algorithm (GA), PSO, and Firefly Algorithm (FA), to predict carbon-steel corrosion rates in Iranian oil-field pipelines. Using 340 samples and variables such as pit depth, exposure time, operational pressure, and chemical concentration, the SVR-GA-PSO model achieved exceptional predictive performance with R^2 of 0.99 and a Root Mean Square Error (RMSE) of 0.0099, outperforming traditional regression-based models. Marins et al. [46] applied several AI models, including RF, ANN, Long Short-Term Memory (LSTM), and Independent Recurrent Neural Network (IndRNN), for fault classification in oil well production systems. Using 1984 temporal well samples characterized by parameters such as tubing and annulus pressure, temperature, and event states, ANN achieved high overall performance (accuracy = 96% and F1-score = 85%). However, the model exhibited limited precision in classifying specific undesirable production events, prompting recommendations for improved hybridization strategies.

In well condition monitoring, Machado et al. [47] deployed One-Class Support Vector Machine (OCSVM) and LSTM models for temporal classification of faults in 3W dataset. Measured variables included tubing and annulus pressure, temperature, and choke pressure. The OCSVM achieved 91% accuracy, effectively identifying early-stage faults despite the limited availability of labeled data. Similarly, Carvalho et al. [48] explored Ordered Nearest Neighbor (ONN) and Weighted Nearest Neighbor (WNN) models to detect flow instabilities in the same well dataset, with ONN reaching an 81% accuracy, suggesting potential for improvement through hybrid metaheuristic tuning.

Classical machine-learning methods, including support vector machines, random forests, and gradient-boosted trees, remain attractive for many surface-network tasks because they perform well on structured tabular data, require less training data than deeper architectures, and offer relatively strong interpretability through feature-importance analysis. In the studies reviewed here, these methods are especially useful for soft sensing, fault classification, integrity screening, and short-horizon prediction. Their main limitation is that they do not naturally capture long temporal dependencies or evolving interactions across interconnected assets, which reduces their robustness for strongly transient multiphase behavior and cross-asset transfer. Accordingly, they are best viewed as strong baselines, practical field-level predictors for moderate-complexity tasks, and useful components within hybrid workflows rather than universal solutions.

3.3. *Artificial neural networks*

Artificial Neural Networks (ANNs) are computational models inspired by the human brain's data-processing mechanisms, capable of learning complex patterns and dependencies through iterative training. Structurally, they consist of input, hidden, and output layers interconnected to mimic biological synapses, enabling the transformation of input data into relevant outputs [49]. Each neuron acts as a processing unit by applying an activation function, such as the sigmoid, tanh, or linear function, while learning algorithms iteratively adjust the connection weights to minimize the error. In the oil and gas sector, ANNs are indispensable for modeling nonlinear systems that traditional statistical or linear approaches cannot handle. Feedforward Artificial Neural Networks (FF-ANNs), which transmit information unidirectionally through hidden layers, are widely adopted for predicting well performance, enhancing production, and assessing pipeline integrity and failure rates [50]. They also serve as efficient surrogate models, replacing computationally intensive simulations for tasks such as predicting pressure drop and estimating slug frequency.

Jia and Zhang [51] used decline curve analysis by training production data with ANNs and time-series methods, achieving a correlation coefficient of 0.99867 for forecasting shale production, exceeding the performance of ANN models trained on bottom-hole pressure (BHP). Building on data-driven strategies, Subrahmanya et al. [52] used active and semi-supervised learning to identify high-information data points in both labeled and unlabeled well datasets. This made production modeling more accurate. In the context of well-performance prediction, Sami and Ibrahim [53] analyzed datasets from Middle Eastern vertical wells using RF, KNN, and ANN models to forecast bottom-hole flowing pressure, with the ANN achieving an R^2 of 93% during testing, highlighting its capability to capture complex nonlinear well behavior. Similarly, Qayyum Chohan et al. [54] trained ANN, Least Squares Boosting (LSB), and Bagging algorithms on 2600 oil shale samples to predict oil yield and CO_2 generation. The ANN model achieved correlation coefficients of 99.6% and 99.9%, respectively, along with low RMSE values, confirming its robustness in modeling multivariable chemical processes.

Ibrahim et al. [55] developed a data-driven framework to classify and predict oil, gas, and water production. Using simulation data from five distinct reservoirs provided by Saudi Aramco, they created a composite dataset containing 12 input features and 1968 data points to

model production behavior over three years. Eight different regression techniques were tested, including Multiple Linear Regression, Polynomial Regression, SVR, Decision Tree Regression, Random Forest Regression, XGBoost, ANN, and Recurrent Neural Networks (RNN), to determine the most accurate predictor of well performance. Their results showed that deep learning methods outperformed traditional ML approaches, achieving high R^2 values of 0.96 for XGBoost, 0.97 for ANN, and 0.98 for RNN, reducing computation time while maintaining strong accuracy in predicting oilfield production rates.

Feed-forward artificial neural networks provide a useful intermediate option between shallow machine-learning models and more complex deep temporal architectures. They are computationally efficient after training and can approximate nonlinear relationships between production variables when the underlying mapping is relatively smooth and close to steady-state. However, because they do not explicitly encode temporal dynamics or surface-network topology, their applicability is more limited for transient forecasting, control-oriented tasks, and multi-node interaction problems. Their predictive quality also depends strongly on data quantity, variable selection, and regularization, which can make them sensitive to overfitting when field data are scarce, noisy, or insufficiently representative.

3.4. Deep learning architectures

Deep Learning (DL) has emerged as a powerful approach for modeling complex and nonlinear multiphase flow phenomena that challenge traditional methods. In the oil and gas sector, DL has transformed analytical practices by enabling advanced pattern recognition and risk mitigation capabilities. Through its multilayered architecture, DL facilitates hierarchical feature extraction, allowing it to effectively capture nonlinear dependencies and handle high-dimensional, temporally dependent data [56]. Architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTMs), and Transformers have demonstrated exceptional performance in modeling temporal dynamics, detecting pipeline anomalies, and predicting operational behavior within production and pipeline systems.

3.4.1. Recurrent neural networks and variants

Since surface network dynamics exhibit strong temporal dependencies, RNNs and their advanced variants, such as LSTM and Gated Recurrent Unit (GRU) models, are highly effective for modeling such behaviors. These architectures capture time-based relationships among flow rates, pressures, and temperatures, enabling accurate forecasting of transient phenomena such as pigging surges and well shut-in responses. Among them, the LSTM network stands out for its ability to preserve long-term dependencies through specialized gating mechanisms, enabling it to accurately predict critical time-dependent parameters such as pressures, flow rates, and production levels essential for dynamic reservoir management. In surface-network applications, these architectures are typically adapted by using sliding time windows of multivariate process variables, such as pressure, temperature, choke position, and flow-rate histories, so that the model can capture transient dependencies associated with slugging, shut-in/restart behavior, gas-lift instability, and short-horizon production fluctuations.

Yang et al. [57] proposed a deep-learning framework using LSTM networks to forecast shale-gas production and bottom-hole pressure with higher fidelity than traditional analytical or statistical models. Their study addressed the limitations of classical decline-curve analysis (Arps, SEPD, and Duong models), which often fail to capture nonlinear dynamics and variable production regimes in unconventional reservoirs. They used a dataset from three shale-gas wells in China's Longmaxi Formation. Historical time-series inputs such as cumulative gas production, pressure, and wellhead conditions were preprocessed before training the LSTM. Their workflow combined short-term statistical forecasting (ARIMA) with a long-term LSTM sequence-learning model that captures temporal dependencies and nonlinear decline behavior. Results indicated that the LSTM achieved remarkably high predictive accuracy, with an R^2 value greater than 0.95 and low mean-square and relative errors, outperforming ARIMA and all traditional decline-curve models. It also generalized across adjacent wells, accurately predicting production at neighboring sites using previously unseen data. However, despite its strong predictive accuracy, the model was evaluated on only three shale-gas wells from a single formation, leaving questions about its scalability and generalization to other reservoir types or production environments. Similarly, Wang et al. [56] analyzed 90,000 samples from the Longmaxi Formation in China's Sichuan Basin to predict real-time natural gas pipeline cracks and production anomalies. Their comparison of Deep-CNN + LSTM, ANN, LSTM, RNN, and SVR models revealed that the hybrid Deep-CNN + LSTM model achieved 99.37% accuracy, underscoring its proficiency at identifying patterns of shale gas production from extensive temporal datasets. In another notable contribution, Antariksa et al. [58] utilized data from the West Natuna Basin (11,497 samples) to predict hydrocarbon production based on well log parameters such as deep and shallow resistivity (LLD, LLS), sonic velocity (Vp), neutron porosity (NPHI), bulk density (RHOB), and gamma ray (GR). Their comparison of LSTM and RF models demonstrated that LSTM achieved a prediction accuracy of up to 94%, with lower RMSE and MSE, highlighting its robustness for temporal gas production forecasting.

Song et al. [59] developed a data-driven oilfield productivity prediction framework using five machine learning algorithms: multivariable linear regression, XGBoost, LGBM, backpropagation (BP) neural network, and LSTM. Their dataset consisted of 394 offshore oil wells from China, featuring 30 parameters including permeability, porosity, oil viscosity, and pressure. After applying Pearson correlation analysis to reduce dimensionality, 12 key features were selected as model inputs. The LGBM model demonstrated the best overall accuracy and generalization, outperforming the XGBoost and LR models, which suffered from overfitting and were sensitive to high-productivity wells. The LSTM model trained on time-series data from a single well showed strong potential for dynamic production forecasting.

Du et al. [60] proposed a Deep Autoregressive Time Series (DATS) model for predicting oil production using U.S. field data from 1919 to 2009. Their approach integrates a modified-Gated Recurrent Unit (M-GRU) network with an autoregressive model to capture both nonlinear temporal dependencies and prior domain knowledge. The M-GRU was designed to incorporate external economic indicators, such as GDP, oil price, and interest rate, alongside technical oilfield parameters. The DATS model achieved the highest predictive accuracy among comparative methods ($R^2 = 0.99874$ and Mean Absolute Error (MAE) = 0.00254), outperforming GRU, LSTM, CNN, and Decline Curve Analysis (DCA) models. It also demonstrated strong performance on small-sample and long-term forecasts, remaining robust during disruptive events such as the 2008 financial crisis and the COVID-19 pandemic. Their model relies on macroeconomic factors rather than detailed reservoir parameters, and therefore, it is less suitable for politically or economically unstable regions, where input data may be uncertain or unavailable.

Martínez and Rocha [61] proposed "The Golem", a general data-driven forecasting model for oil and gas production based on RNNs, specifically using LSTM and GRU architectures. Their model aimed to overcome the limitations of prior work, which focused solely on single-variable predictions (typically oil rate) and a single class of reservoir data. They developed an adaptive sequence-to-sequence (Seq2Seq) forecasting framework capable of predicting multiple production variables (oil, gas, water rates) and well-bore pressure. They utilized two datasets: the UNISIM-II-H benchmark (a synthetic carbonate reservoir from the Brazilian pre-salt) and the Volve field (a

decommissioned North Sea oilfield). The model ingested daily historical production data and generated forecasts for variable-length future time horizons, with its structure automatically adjusting to dataset size and time window. Key findings showed that GRU and LSTM models achieved comparable performance, often reaching R^2 values of up to 0.99 and Symmetric Mean Absolute Percentage Error (SMAPE) below 2% in synthetic cases, while still maintaining a strong trend-following ability for real-world data. The model's simplicity, employing minimal dense layers and no physics-based inputs, demonstrated high efficiency and adaptability across different production scenarios. However, the model's accuracy decreased in real-field applications (e.g., Volve) due to data complexity and shut-in periods, and it lacked integration with physics-based constraints, which may limit interpretability and extrapolation to unseen conditions.

Werneck et al. [62] developed a data-driven deep learning framework for short-term forecasting of oil production and bottom-hole pressure in complex reservoirs. Their dataset included sensor data from producers and injector wells, including oil, gas, water production, and pressure measurements. They tested several neural network architectures: stacked GRU, CNN, Seq2Seq, and hybrid GRU–CNN models, against both conventional methods (Pressure-Normalized Decline Curve Analysis) and modern off-the-shelf forecasting models such as Transformer, N-BEATS, DeepAR, and Prophet. Simulations were performed on both real field data and a synthetic benchmark dataset (UNISIM-II-M-CO), demonstrating that stacked recurrent networks (GRU and Seq2Seq) achieved the best performance for 30-day forecasts of production rates and pressure. Their approach also incorporated anomaly removal and data augmentation to improve model robustness.

3.4.2. Transformers

Transformer models have emerged as one of the most influential architectures in modern AI, fundamentally transforming how sequential and structured data are processed. Introduced by Vaswani et al. [63], the Transformer model replaced recurrent and convolutional operations with self-attention mechanisms that capture long-range dependencies in data more efficiently and in parallel, thereby drastically improving scalability and performance across tasks. Its encoder–decoder framework, multi-head attention, and positional encodings allow it to learn complex temporal and spatial relationships within sequences. Beyond natural language processing (NLP), the Transformer's ability to model non-linear, multivariate dependencies makes it particularly suitable for time-series forecasting and dynamic system modeling. Recent cross-domain studies have further strengthened this view by demonstrating Transformer-based advances for multi-granularity spatiotemporal forecasting, non-stationary sequence learning, and robust multivariate forecasting with missing values [64,65]. In the oil and gas industry, Transformer architectures can be leveraged to predict production rates, optimize surface network operations, and detect anomalies by learning intricate dependencies among wellhead pressures, flow rates, and reservoir conditions. Li et al. [66] introduced TST-Refrac, a novel Transformer-based model designed to predict post-fracturing production in re-fractured oil wells. Using time-series production data from Block W in the Junggar Basin, the authors aimed to overcome the limitations of traditional deep learning models, such as RNNs and LSTMs, which struggle with long-range dependencies and non-stationary production data. Their approach employed a Transformer encoder architecture enhanced with a sliding-window method to capture temporal dependencies. The model was trained and validated on production data from nine wells and benchmarked against RNN and LSTM models. TST-Refrac achieved an average RMSE of 0.4475, significantly outperforming the RNN (1.1053) and LSTM (0.8806). However, the study's limitations include a small sample size (nine wells for model training and six for validation), a limited geographic scope, and the model's reliance solely on production-rate time series, without incorporating operational or pressure-related features, which may limit its broader applicability and robustness across diverse reservoir conditions.

Huang et al. [67] developed a Time Patch Dynamic Attention Transformer (TPDAT) model to enhance well production forecasting in complex, noisy oilfield environments. Their work addressed the weaknesses of traditional deep-learning models: LSTM's vanishing gradients on long sequences, CNN's limited receptive fields, and the standard Transformer's inability to preserve causality and local temporal features. Using the Volve oilfield dataset (2008–2016) from the North Sea, the authors selected three production wells. Input variables included downhole and tubing pressures, choke settings, wellhead pressures, temperatures, and water production. The model's performance, benchmarked against CNN, LSTM, Temporal Convolutional Network (TCN), and standard Transformer baselines, showed substantial improvements. However, while TPDAT demonstrated excellent accuracy and generalization, its complex model architecture and limited testing on a single dataset (Volve field) restrict its generalizability across other reservoir types. Moreover, the model's reliance on intensive hyperparameter tuning and lack of interpretability in physical terms remain challenges for its deployment in real-time or physics-guided operational workflows.

3.4.3. Graph neural networks

Graph Neural Networks (GNNs) represent an emerging class of deep learning architectures specifically designed to model systems with inherent relational and topological structures [68]. Unlike traditional neural networks that operate on Euclidean data (e.g., grids or sequences), GNNs can learn from graph-structured data, where nodes represent entities and edges encode their spatial, physical, or functional relationships. Recent studies summarized by Corradini et al. [69] highlight how spatio-temporal GNNs extend classical GNNs by jointly capturing spatial interactions and temporal evolution in multivariate time series data. These models integrate convolutional, recurrent, and attention-based modules to learn dependencies among multiple variables across space and time, rendering them well-suited for dynamic production environments.

In oil and gas surface networks, GNNs can model the interconnected relationships among wells, manifolds, separators, and sensors, where nodes represent system components and edges capture physical flowlines or operational dependencies. GNNs can accurately predict production rates, pressure changes, and equipment performance under different operating conditions by combining process topology with data-driven learning. Beyond prediction, GNNs enable network-level optimization by learning how control actions at one point in the system affect others, supporting real-time optimization, anomaly detection, and fault localization. Integrated within hybrid or physics-informed frameworks, they can serve as the core of digital twins, enhancing scenario analysis, adaptive control, and overall production efficiency in complex surface networks.

Lijie et al. [70] presented a GNN-based framework for identifying accident chains in oil and gas pipeline systems. The proposed GNN model represented pipeline components (e.g., valves, sensors, and sections) as nodes and their interdependencies as edges, thereby modeling the causal pathways leading to accidents. The framework integrated structured data (955 production anomaly records) and unstructured data (257 detailed accident reports) to construct semantic and causal graphs. Using techniques such as semantic segmentation, dependency parsing, and attention-based GNNs, the model extracted key elements from text data and built co-occurrence networks to infer accident chains. These chains revealed critical risk propagation patterns, for example, how corrosion, valve malfunctions, or delayed communication could cascade into fire or explosion events. Results showed that the GNN framework accurately identified high-risk nodes and causal sequences, improving predictive accuracy and interpretability compared to

conventional methods. It highlighted corrosion and human error as dominant contributors to cascading failures, providing a basis for preventive maintenance, real-time monitoring, and safety optimization in pipeline systems.

3.4.4. Multi-agent systems

Multi-Agent Systems (MAS) are computational frameworks comprising multiple autonomous agents that interact within a shared environment to achieve common objectives [71]. Each agent perceives its surroundings, interprets information, and selects actions to optimize performance toward specific goals. MAS architectures enable decentralized, adaptive, and collaborative control, rendering them highly effective for complex industrial applications, such as oil and gas operations, where decision-making involves numerous dynamic, interdependent variables. Their distributed structure allows multiple agents to operate in parallel, managing localized control or optimization tasks while contributing to overarching system objectives, an approach well-suited to O&G environments with variable conditions, evolving field dynamics, and competing priorities across production, maintenance, and supply chain management.

To enhance oil output and reduce unwanted water and sand production, Engmo and Hallen [72] developed a MAS framework for production control and optimization. Using the Prometheus design methodology [73] and the JACK intelligent agent environment [74], they designed a distributed architecture comprising agents responsible for well monitoring, processing plant supervision, and operator coordination. Each agent analyzed real-time sensor data, such as permeability and water-to-sand ratios, and autonomously adjusted control strategies. The system's performance was tested using a simulated oil field model built in the Eclipse environment. When compared with manual control, the MAS achieved higher oil production rates and faster response times, demonstrating superior adaptability under critical operational conditions. The study concluded that MAS can effectively manage specific supervisory tasks in production systems, reducing human workload while maintaining higher efficiency. Building on this, Mikkelsen et al. [75] introduced a multi-objective hierarchical MAS for offshore oil and gas production control. Their system was structured across three layers: strategic, tactical, and operational, with agents negotiating objectives within a shared decision-making framework. Using data from DONG Energy EP-operated Siri Field in the North Sea, each agent leveraged historical and real-time production data to optimize parameters related to flow rates, pressure constraints, and environmental limits. Through iterative negotiation, agents collaboratively balanced competing objectives, resulting in enhanced operational efficiency, reduced environmental impact, and improved economic performance compared to traditional manual systems. In another contribution, Dobrescu et al. [76] proposed a distributed MAS control framework for onshore petroleum production systems. The model transformed conventional control units into autonomous production unit agents capable of executing control functions as modular services. The system's distributed architecture ensured dynamic load balancing and scalability, enabling better adaptability to varying production demands. Although the framework was validated through simulation, the authors emphasized its potential for real-world deployment in managing onshore field operations under uncertain and fluctuating conditions. These studies demonstrate that multi-agent systems offer a practical, decentralized approach for managing the complex, nonlinear, and time-varying processes inherent to O&G production. By facilitating autonomous decision-making, real-time collaboration, and adaptive control, MAS architecture contributes to optimized production performance, enhanced responsiveness, and greater operational reliability in both onshore and offshore environments.

Deep sequence, attention-based, and graph-based architectures offer the strongest potential for complex transient forecasting and topology-aware modeling because they can represent long-range temporal dependencies and interactions among multiple assets. In the reviewed studies, recurrent models and Transformer-based models generally show stronger performance than simpler methods for multivariate forecasting, while graph-based approaches are conceptually well matched to field gathering systems and interconnected production networks. Their main trade-offs are higher data and computational demands, greater implementation complexity, and lower operator-facing interpretability. In addition, much of the reported performance remains dataset-specific, with limited cross-asset validation and relatively few large-scale field deployment studies, which makes generalizability an ongoing concern.

3.5. Hybrid and physics-informed machine learning models

3.5.1. Hybrid gray-box models

Hybrid (gray-box) modeling combines mechanistic equations with machine learning corrections to capture complex system behavior more accurately than either approach alone. A common strategy is residual learning, where an ML model learns the discrepancy between predictions from a physics-based simulator and observed data. This structure has been particularly effective in VFM, where a physics-based flow model provides baseline predictions that are dynamically corrected by a data-driven component trained on real-time sensor inputs. Such synergy enhances accuracy, robustness, and generalization across varying flow conditions, while reducing dependence on extensive labeled datasets. In practical surface-network settings, this is often implemented by using a mechanistic simulator or reduced-order flow model to represent the dominant hydraulic behavior, while a learned residual model corrects for unmodeled effects such as changing fluid properties, sensor bias, flow-regime transitions, or asset-specific mismatch. Shi et al. [77] proposed a Hybrid Physics-Guided Variational Bayesian spatial–temporal neural network that integrates physical parameters (e.g., geometry, release rate, sensor positions) with deep learning for spatiotemporal gas forecasting. The model achieved an R^2 of 0.99 using 600 temporal samples, demonstrating the strong complementarity between physics-informed priors and data-driven inference in capturing the dynamic behavior of the field. Souza et al. [78] proposed a hybrid modeling approach to predict the nonlinear dynamics of a gas-lift oil production system. Their method combined a first-order dynamic model with variable gain to represent the system's base response and an encoder–decoder GRU neural network to model the residual, nonlinear component. Using simulated data from a gas-lift well, they evaluated the approach under both open- and closed-loop conditions, including the presence of noise and unmeasured disturbances. The hybrid model achieved normalized root mean squared errors below 5% and stationary residuals between 90% and 100%, outperforming a direct GRU model without decomposition, which yielded only 83% stationary residuals and a tenfold higher training error. Similarly, Dong et al. [79] developed a hybrid neural network model that combines fully connected (dense) layers and LSTM layers to improve shale gas production prediction, addressing delays in single-step prediction and enabling its application to new wells. They used daily production data from over 300 shale gas wells encompassing more than 200,000 production days, incorporating gas rate, water rate, production time, and manually controlled shut-in periods as input features. Their model introduced feature engineering steps that added produced water, bottom-hole total flow rate, and production plan variables (e.g., shut-in duration and cumulative production times) to enhance temporal and operational context. Results showed that the proposed model significantly outperformed traditional LSTM and decline curve analysis models. For a 14-day prediction window, the mean cumulative error was just 40.26% of that observed with a standard LSTM model. Over a 100-day window, the error represented 77.93% of the LSTMs and only 29.71% of the classic Arps DCA models.

3.5.2. Physics-informed learning

The modern era of production optimization in oil and gas systems is increasingly shaped by Physics-Informed Machine Learning (PIML). This approach unites data-driven learning with fundamental physical

principles. Traditional ML methods excel at pattern recognition but often lack physical interpretability and perform poorly outside their training range. At the same time, purely physics-based models, though rigorous, can be computationally expensive and limited by simplifying assumptions. PIML bridges this gap by embedding domain knowledge directly into the learning process, yielding fast, differentiable, and physically consistent surrogate models that combine the robustness of physical modeling with the adaptability of ML. A prominent realization of this paradigm is the Physics-Informed Neural Network (PINN), introduced by Raissi et al. [28]. PINNs integrate governing equations, typically partial differential equations (PDEs), into a neural network's loss function through a physics-based penalty term that enforces conservation laws. For surface-network applications, these constraints may include mass, momentum, and energy balances, together with boundary and initial conditions derived from wellhead measurements, separator conditions, or pipeline operating limits. In this way, PINNs are not trained only to fit observations, but also to remain consistent with the underlying multiphase-flow physics when measurements are sparse, noisy, or only partially available. By leveraging automatic differentiation to compute PDE residuals, they can learn from sparse or operationally noisy data while maintaining physical consistency. This reduces reliance on large, labeled datasets and mitigates the risk of unrealistic predictions common in purely data-driven models. In surface network applications, PINNs can accurately capture pressure and temperature distributions in multiphase pipelines, providing a natural framework for hybrid modeling that combines limited sensor data with partial physical models. By preserving mass, momentum, and energy conservation, they deliver improved extrapolation capability, stability, and reliability, making them particularly suitable for integration into digital twin frameworks for real-time production monitoring, forecasting, and optimization.

Faria et al. [80] developed a hybrid framework combining PINNs and Deep-Reinforcement Learning (DRL) to optimize oil production in gas-lifted well networks. They utilized PINNs to model the nonlinear, multiphase flow dynamics of each well by incorporating governing physical equations into the neural network's loss function. This made it less reliant on large operational datasets. The PINN models served as a fast, physics-consistent simulation environment for training a DRL agent that optimized gas injection rates for economic performance. Their case study involved a three-well network with process data derived from prior validated dynamic models. Results showed that the PINN-based simulator accurately reproduced the behavior of differential and algebraic process states, even under noise, and that a DRL agent trained on it achieved comparable oil production and policy performance to a nonlinear model predictive controller while outperforming a DRL agent trained on differential–algebraic equation models.

In another work, Kittelsen et al. [81] developed an improved Physics-Informed Neural Network for Control (PINC) framework for modeling and controlling highly nonlinear systems, specifically gas-lifted oil wells. They enhanced the original PINC method by introducing skip connections to prevent vanishing gradients and by modifying problematic terms in the governing ODEs (e.g., logarithmic and square-root terms) to ensure stable training. Their hierarchical architecture combined a PINC module that predicts system states with a secondary neural network that estimates algebraic variables, such as bottom-hole pressure. Using simulation data from a differential-algebraic model of gas-lifted wells, they trained and validated the networks against Runge–Kutta benchmarks. The improved PINC achieved a 67% reduction in prediction error and increased gradient magnitudes by four orders of magnitude compared to the baseline. At the same time, nonlinear model predictive control tests demonstrated the accurate and robust regulation of bottom-hole pressure, even under noisy measurement conditions.

3.5.3. Neural operator learning

Neural Operator (NO) learning represents the next significant advancement in physics-informed modeling, extending beyond the instance-specific nature of PINNs by learning mappings between entire function spaces. Unlike traditional neural networks that approximate discrete solutions, NOs can model families of PDEs under varying boundary conditions and parameters without retraining. This makes neural operators particularly attractive for repeated scenario analysis in pipeline and surface-network systems, where operating conditions, boundary conditions, and control settings change continuously. In oil and gas applications, their practical value lies in learning fast surrogate mappings from network states and operating inputs to pressure, temperature, or flow responses, thereby accelerating optimization and control loops that would otherwise rely on expensive transient simulation. This capability also makes them effectively resolution-invariant, meaning that models trained on coarse data can often predict accurately at finer spatial or temporal scales, an especially valuable property for large, dynamic systems such as offshore production networks.

Among NO architectures, the Fourier Neural Operator (FNO) has emerged as a leading approach. By parameterizing integral kernels in Fourier space and leveraging Fast Fourier Transform (FFT) operations, FNO efficiently captures global spatial dependencies, achieving orders of magnitude faster computation than traditional numerical solvers for turbulent or multiphase flows. Ma et al. [82] employed FNO to efficiently simulate subsurface multiphase (oil–water) flow in heterogeneous reservoirs. They formulated the governing partial differential equations for two-phase flow and used FNO to learn mappings between permeability–porosity fields and reservoir pressure, leveraging spectral information through Fourier transforms. Their dataset comprised 1000 synthetic 2D reservoir realizations (241×241 grids) generated using a Gaussian simulator, with the results compared to CMG IMEX simulations. The model achieved high accuracy (low MAE and RMSE) and demonstrated zero-shot super-resolution capability, predicting high-resolution (1205×1205) pressure fields without retraining. However, accuracy decreased near wells due to abrupt pressure gradients, which they mitigated using a multi-fidelity approach that combined low- and high-resolution data (MF20, MF50 models). This refinement further reduced prediction errors and maintained visual consistency with simulator outputs. In another work, Gong et al. [83] developed an efficient model for simulating transient processes in natural gas pipelines using an FNO framework enhanced with a physics-based constraint. Their method, termed FNO-BWRS, embedded the Benedict-Webb-Rubin-Starling (BWRS) equation of state into the model's loss function to integrate thermodynamic principles and improve prediction accuracy. The model was trained on synthetic data generated by a self-developed numerical solver simulating 120 hours of flow in a 100 km pipeline, incorporating hydraulic, thermal, and compositional variables. Results showed that the FNO-BWRS model achieved root mean square and mean absolute errors on the order of 10^{-3} with $R^2 > 0.99$, outperforming the baseline FNO and a U-Net convolutional model. Their model also demonstrated strong grid invariance and achieved computational speeds significantly higher than those of traditional numerical solvers, enabling rapid, accurate, and generalizable transient pipeline simulations.

Building on this foundation, the Physics-Informed Neural Operator (PINO) incorporates physical constraints directly into the operator learning process, combining the generalization strength of FNOs with the physical consistency of PINNs. This integration yields improved convergence and accuracy for long-horizon transient simulations in pipelines and production networks, while requiring less data and exhibiting superior generalization performance. A related development, the Deep Operator Network (DeepONet), learns mappings between input and output function spaces, allowing fast inference without repeatedly solving the governing PDEs. DeepONet architectures can

emulate high-fidelity multiphase flow simulators using representative training datasets and generalize to unseen boundary or initial conditions. In the oil and gas industry, these operator-learning frameworks are particularly valuable for surface network modeling and optimization, enabling rapid scenario testing for pipeline transients, hydrate formation prediction, and adaptive control of gas-lift or choke systems. By learning the governing operators of flow and transport directly, NOs, PINOs, and DeepONets provide a robust computational foundation for real-time digital twins, offering the combined benefits of physical fidelity, computational efficiency, and scalability for production forecasting and optimization.

Hybrid gray-box models, physics-informed neural networks, and neural-operator approaches directly address one of the central weaknesses of purely data-driven AI, namely weak extrapolation outside the training domain. By embedding physical structure, conservation constraints, or simulator knowledge, these methods offer a more defensible route for safety-relevant oil and gas applications. The reviewed studies suggest that such approaches can improve robustness, reduce dependence on fully labeled datasets, and accelerate scenario evaluation relative to high-fidelity first-principles simulators. Their present limitations are the higher effort required for formulation, calibration, and loss balancing, as well as a still-limited body of field-scale validation under unstable multiphase and rapidly changing operating conditions. Taken together, these studies indicate that physics-informed models can remain reliable under noisy measurements and changing boundary conditions, but evidence is still limited for abrupt flow-regime transitions, severe slugging, and sustained field-scale disturbances in fully operational multiphase systems.

3.6. Optimization algorithms and generative AI

Optimization remains a cornerstone of production network management, aiming to balance throughput maximization, cost efficiency, and operational risk reduction. Evolutionary algorithms, particularly genetic algorithm (GA) and PSO, have proven highly effective for addressing nonlinear and multidimensional problems, such as those in oil and gas systems [84]. GA employs evolutionary strategies to optimize complex network configurations, such as pipeline diameters, compressor placement, and operating pressures. Its ability to escape local minima makes it especially suitable for large, nonconvex optimization problems. It has been successfully applied to field development planning, production scheduling, reservoir characterization, and seismic inversion, demonstrating broad applicability across the upstream value chain [85]. In surface network operations, PSO can be leveraged for tasks such as production allocation, real-time choke setting optimization, and pigging schedule planning. Hybrid approaches that couple GA or PSO with machine–learning–based surrogate models can significantly improve computational efficiency and convergence speed [86,87]. These hybrid optimization strategies enable rapid scenario evaluation and adaptive decision-making, supporting the continuous optimization of complex, data-driven production networks.

Recent advances in Generative AI have also transformed the optimization landscape. Generative Adversarial Networks (GANs) [88] and Variational Autoencoders (VAEs) [89] can augment scarce field data by generating synthetic flow scenarios, improving model robustness. Moreover, Large Language Models (LLMs) [90] fine-tuned for engineering contexts are increasingly used for automated report generation, operational summarization, time-series forecasting, and the recommendation of optimization strategies. Together, these approaches enable data augmentation, scenario simulation, and decision automation, paving the way for intelligent, self-optimizing production systems.

Optimization algorithms and generative-AI methods broaden the role of AI/ML from prediction toward decision support, scenario generation, and automated operational assistance. Evolutionary optimizers are valuable for nonlinear and nonconvex production problems, especially when coupled with surrogate models, whereas generative methods can help address data scarcity and broaden scenario coverage. However, their usefulness in production operations depends strongly on constraint handling, operational safeguards, and integration with human oversight.

3.7. Summary

This section presented a comprehensive overview of AI and ML applications in surface network modeling, prediction, and optimization for oil and gas production systems. It highlighted the transition from traditional physics-based simulators to data-driven approaches capable of learning complex nonlinear dynamics and providing real-time operational understandings. Key data sources, such as pressure, temperature, flow rates, choke settings, and fluid properties, were identified as essential for training and validating AI/ML models. The section reviewed a range of statistical, machine learning, and deep learning methods, including regression, tree ensembles, support vector machines, and neural network architectures (ANNs, LSTMs, CNNs, and Transformers), with an emphasis on their roles in production forecasting, fault detection, and anomaly classification. The section further explored emerging paradigms, including multi-agent systems, hybrid gray-box modeling, and physics-informed learning, which integrate physical principles with machine learning to enhance model accuracy, interpretability, and scalability. Recent advancements in neural operator learning, evolutionary optimization algorithms (GA, PSO), and generative AI were also highlighted as enablers of adaptive, self-optimizing digital twins.

Table 3 summarizes recent advancements in AI/ML for oil and gas production. Sequence models such as LSTMs and hybrid approaches consistently deliver strong performance for time-series forecasting, while ensemble methods like XGBoost excel at classification. However, limitations such as small sample sizes, data quality issues, and lack of field validation remain common. Future work should focus on improving data curation, transferability, and interpretability to enable broader industrial adoption.

Although a substantial portion of the reviewed literature remains simulation-based, several studies already provide field-based or operationally relevant evidence of practical deployment potential. For example, Transformer-based production forecasting was evaluated on the Volve field using well-level pressure, choke, and temperature data from three production wells [67], while related multi-variable forecasting studies also tested recurrent models on real-field Volve data in addition to synthetic benchmarks [61]. Field-based forecasting is also reflected in the Block W post-fracturing case [66] and in hybrid forecasting across 300 shale-gas wells, which supports pad-level operational use rather than only laboratory-scale demonstration [79]. For production assurance and event recognition, the Petrobras 3W benchmark has been widely used for classifying undesirable operating conditions from real production variables [41,42]. At the control and optimization level, the hierarchical multi-agent study on the Siri Field illustrates how AI-assisted decision-making can be aligned with offshore operational objectives using historical and real-time production information [75]. Taken together, these examples show that industrial relevance is increasing, but they also confirm that most current implementations remain asset-specific and short of fully validated, field-wide closed-loop deployment.

From an operational perspective, the reviewed studies also suggest that AI/ML value depends on asset context. In offshore environments, where logistics are constrained and response times are critical, approaches such as hierarchical multi-agent control and edge-enabled monitoring are particularly attractive because they support faster decision-making, reduced operator burden, and improved coordination across distributed assets. In onshore systems, where scalability and field-wide integration are often more important, distributed control frameworks offer advantages for handling variable production demands across larger operational footprints. The benefits are especially relevant

Table 3
Overview of AI/ML modeling approaches applied to surface network production systems.

Reference	Category	Model/ Algorithm	Objective	Dataset (Type/Size)	Key Features	Performance Metric	Application Readiness	Limitations
Jia & Zhang, 2016 [51]	Time-Series Forecasting	ANN (decline-curve assisted)	Shale production forecasting	Historical production data	Decline features, time-series	$R^2 = 0.99867$	Baseline production forecasts	May not capture interventions
Qiao et al., 2017 [35]	Kernel Methods	PSO–LS-SVM	Short-term oil & gas forecasting	Field, Temporal	Cleaned time series, auto hyperparameters	High convergence speed/precision	Rate/pressure predictions for allocation	Reliability drops for long-horizon
Panja et al., 2018 [36]	Surrogates & Kernel	RSM, LS-SVM, ANN	Hydrocarbon recovery & GOR response	Synthetic, non-temporal	Permeability, GOR, compressibility, BHP, etc.	RSM & LS-SVM are strong for oil recovery	Scenario screening, planning	Simulated data, limited physics
Liu et al., 2021 [38]	Ensembles	ANN, LR, SVM, GPR, DT	Pipeline defect prediction	Field logs: 1500 samples	Well/pipeline log attributes	ANN $R^2 \approx 0.99$ train / 0.96 test	Pre-emptive integrity interventions	Possible overfit, explainability needed
Yin et al., 2021 [44]	Failure Analysis	MLP, RF, SVR	Pipeline failure analysis	Field: 149,940 records, non-temporal	Operational, safety, maintenance	MLP & SVR are competitive	Large-scale reliability monitoring	Label noise, heterogeneous sources
Marins et al., 2021 [46]	Fault Classification	RF, ANN, LSTM, IndRNN	Fault classification in production	1984 samples, Temporal	Tubing/annulus pressures and temperatures, event states	ANN accuracy ≈ 96%	Event classification for mitigation	Lower precision on rare events
Sami & Ibrahim, 2021 [53]	Regression	ANN vs. RF, KNN	BHP prediction	Field wells, non-temporal	Well parameters, pressure	ANN test $R^2 \approx 0.93$	BHP proxy for choke optimization	Generalization across fields uncertain
Carvalho et al., 2021 [48]	KNN Variants	ONN, WNN	Flow instability detection	3W well dataset, Temporal	P, T, choke settings	ONN ≈ 81% accuracy	Instability alarms for protection	Room for improvement via metaheuristics
Kumar & Hassanzadeh, 2021 [39]	Random Forest	RF	Reservoir barrier performance	Synthetic, Temporal	Formation compressibility, thermal cond.	High R^2, low RMSE	Scenario screening for constraints	Synthetic data
Santos et al., 2021 [41]	Anomaly Detection	RF	Well integrity/ production anomaly	Temporal well data	Gas-lift, DSCP, flowrates	Accuracy ≈ 94%	Real-time anomaly flagging	Event class imbalance, alarm fatigue risk
Machado et al., 2022 [47]	Fault Detection	OCSVM, LSTM	Early-stage fault detection	1984 samples, Temporal	P, T, DSCP	OCSVM ≈ 91% accuracy	Unsupervised anomaly screening	Sparse labels, tuning sensitivity
Qayyum Chohan et al., 2022 [54]	Process Modeling	ANN, LSB, Bagging	Oil yield & CO_2 generation	Lab: 2600 samples, non-temporal	Geochemical/ processing variables	$R^2 \approx 99.6\%$ (yield)	Refines compositional models	Lab-to-field transferability limited
Ibrahim et al., 2022 [55]	Rate Prediction	XGBoost, ANN, RNN	Oil, gas, and water rate prediction	Simulation: 1968 rows, 12 features	Engineered features over 3 years	R^2: XGBoost ≈ 0.96, ANN ≈ 0.97	Benchmark for algorithm choice	Simulator bias, needs field adaptation
Wang et al., 2022 [56]	Anomaly Detection	Deep-CNN + LSTM	Pipeline crack & anomaly detection	Field: 90,000 samples, Temporal	Temporal production signals	Accuracy ≈ 99.37%	High-recall anomaly screening	Class imbalance, explainability
Yang et al., 2022 [57]	Short-Term Forecasting	LSTM + ARIMA	Shale-gas production & BHP forecasting	Field: 3 wells, Temporal	Cumulative gas, P, well head conditions	$R^2 > 0.95$	Surface network response forecasting	Small sample, formation-specific
Werneck et al., 2022 [62]	RNN Stacks & Hybrids	GRU, CNN, Seq2Seq, GRU–CNN	30-day forecasts of rates & BHP	Field + synthetic, Temporal	Producer/injector rates & pressure	Stacked GRU & Seq2Seq best	Short-term forecasting for operations	Needs anomaly removal & augmentation
Ma et al., 2023 [7]	Tree Ensembles	RF, XGBoost, SVM, LGBM	Burst pressure estimation (corroded pipe)	Field: 314 samples, non-temporal	Geometry, material strength, corrosion	$R^2 \approx 99\%$ train, 98% test	Constraint handling in optimization	Small sample, domain shift risk
Fang et al., 2023 [43]	Gradient Boosting	CatBoost vs. RF	Pipeline corrosion rate prediction	Field: 3240 samples, non-temporal	Gas composition, pressure, velocity	Best accuracy among tested models	Predict corrosion for pigging/ chemicals	Needs periodic calibration
Naserzadeh & Nohegar, 2023 [45]	SVR (Metaheuristic)	SVR-GA-PSO	Carbon steel corrosion rate prediction	Field: 340 samples, non-temporal	Pit depth, exposure time, chemical conc.	$R^2 \approx 0.99$, RMSE ≈ 0.0099	Chemistry set-point optimization	Small dataset, overfitting risk
Antariksa et al., 2023 [58]	Well Forecasting	LSTM vs. RF	Gas production forecasting	Field: 11,497 samples, Temporal	Well logs, petrophysical features	Accuracy ≈ 94%	Well-level rate forecasts	Feature scaling & QC critical
Song et al., 2023 [59]	Productivity Prediction	LGBM, XGBoost, LSTM	Offshore productivity prediction	Field: 394 wells, 12 features	Permeability, porosity, and pressures	LGBM best overall	Screening & per-well TS forecasting	Overfit risk for high-productivity wells
Martínez & Rocha, 2023 [61]	Multi-Variable Forecast	LSTM/GRU Seq2Seq	Production & pressure forecasting	Synthetic & field datasets, Temporal	Daily rates, wellbore pressure	$R^2 \approx 0.99$ (synthetic)	General forecasting engine	Accuracy drops on real field, lacks physics
Du et al., 2024 [60]	Long-Horizon Forecast	DATS (GRU + AR)	Oil production forecasting	US field data, Temporal	Macro indicators + technical features	$R^2 \approx 0.99874$, MAE ≈ 0.00254	Macro-level planning	Relies on macro data, less suitable in unstable regions

(*continued on next page*)

Table 3 (*continued*)

Reference	Category	Model/ Algorithm	Objective	Dataset (Type/Size)	Key Features	Performance Metric	Application Readiness	Limitations
Li et al., 2024 [66]	Transformer	TST-Refrac	Post-fracturing production prediction	Field: Block W, Temporal	Production-rate time series	Avg. RMSE ≈ 0.4475	Specialist predictor for re-frac planning	Small sample, lacks operational features
Huang et al., 2024 [67]	Transformers	TPDAT	Well production forecasting	Field Volve: 3 wells, Temporal	Downhole/ tubing P, choke size, Well head P & T	TPDAT performed better than other models	Advanced forecasting in noisy settings	Complex architecture, tuning, interpretability
E. Souza et al., 2024 [78]	Hybrid Gray-Box	First-order + GRU	Gas-lift oil-well dynamics	Simulated gas-lift well, Temporal	Base model + GRU residual	Normalized RMSE < 5%	Well-level VFM/ soft sensor	Simulation only, needs field validation
Dong et al., 2025 [79]	Hybrid Gray-Box	Dense + LSTM	Shale gas production prediction	Field: 300 wells, Temporal	Gas/water rate, engineered variables	MAE 14d ≈ 40% of LSTM	Operational forecasting across pads	Manual plan variables required

Legend: P = Pressure, T = Temperature.

Table 4
Comparative trade-off matrix of major AI/ML families for oil and gas surface networks.

Model family	Best-suited tasks	Main strengths	Main limitations	Data demand	Interpretability / physical consistency	Deployment readiness
Statistical and classical ML	Soft sensing, integrity screening, short-horizon prediction, anomaly classification	Data-efficient, fast training/ inference, comparatively interpretable	Limited for long transients, weak topology awareness, performance degrades under distribution shift	Low to medium	Medium to high	High
Tree ensembles (RF, XGBoost, LGBM)	Tabular production prediction, classification, fault detection	Strong baseline performance, robust on mixed tabular features, native feature-importance measures	Limited temporal memory, not physics-aware, less suitable for dynamic network interactions	Low to medium	Medium to high	High
Feed-forward ANN	Steady-state or weakly dynamic production prediction	Good nonlinear approximation, computationally efficient after training	No native temporal or graph structure, overfitting risk on limited field data	Medium	Low to medium	Medium to high
Recurrent deep learning (RNN/ LSTM/GRU/TCN)	Multivariate time-series forecasting, virtual flow metering, transient prediction	Captures temporal dependence, strong forecasting capability	More data-hungry, harder to interpret, weaker transferability across assets	Medium to high	Low	Medium
Transformer / attention models	Long-horizon forecasting, multivariate sequence modeling under complex dependencies	Strong sequence modeling, handles long-range interactions better than many recurrent models	High computational cost, complex tuning, limited interpretability	High	Low to medium	Medium
Graph neural networks	Network-scale prediction, topology-aware monitoring and optimization	Explicitly represents field connectivity and asset interactions	Requires graph construction and maintenance, limited field-scale benchmarking	High	Medium	Medium
Hybrid gray-box models	Forecasting and control where partial physics is available	Combines plant knowledge with data-driven flexibility, better robustness than purely data-driven models	Requires careful architecture design and calibration, depends on physics-model quality	Medium	High	Medium to high
Physics-informed neural networks	State estimation, constrained forecasting, physics-consistent surrogates	Improves physical plausibility, better extrapolation potential, reduced reliance on fully labeled data	Sensitive to loss balancing and formulation choices, limited large-scale field validation	Medium	High	Medium
Neural operators (FNO, PINO, DeepONet)	Fast surrogate simulation, repeated scenario analysis, operator learning	Promising speed and scalability for complex physics-driven mappings	Emerging for field deployment, demanding implementation and validation requirements	High	Medium to high	Low to medium
Optimization / generative-AI layers	Scenario generation, decision support, surrogate-assisted optimization	Useful for nonlinear search, planning, and data augmentation	Requires guardrails, constraint enforcement, and human oversight for operational use	Variable	Variable	Low to medium

for mature assets, where AI/ML can help preserve throughput, reduce unplanned downtime, and improve operational efficiency under declining reservoir conditions and evolving network constraints. Where quantified in the reviewed literature, reported gains include higher oil production rates, faster response times, and improved operational efficiency, although direct cost-reduction metrics remain inconsistently reported across studies.

Across the reviewed literature, no single AI/ML family is uniformly optimal for all surface-network tasks. Instead, the most suitable choice depends on the balance required among predictive performance, temporal or topological fidelity, data availability, interpretability, physical consistency, and deployment constraints. While Table 3 summarizes representative studies and their reported outcomes, direct numerical comparison across the literature remains difficult because the reviewed works differ in datasets, target variables, forecast horizons, and validation settings. To provide a more standardized basis for comparison, Table 4 summarizes the major AI/ML families reviewed in this paper using common dimensions relevant to oil and gas surface networks. This comparison highlights that model selection should be guided by the intended operational use case and the practical constraints of implementation rather than by predictive performance alone.

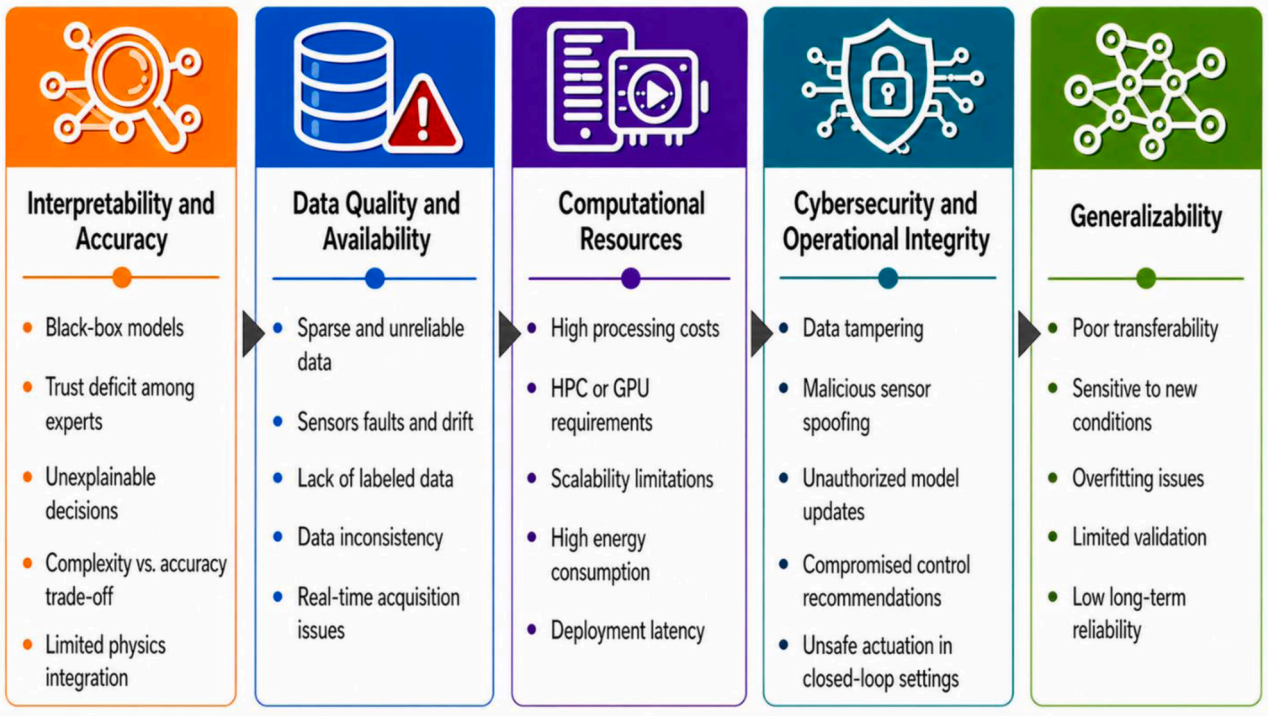


Fig. 7. Overview of significant challenges limiting AI/ML adoption in surface network optimization.

4. Challenges and limitations

While AI/ML applications in surface network optimization have shown promising results, their widespread adoption in oil and gas operations is constrained by several critical challenges. These limitations pertain to interpretability, data quality, computational feasibility, and the ability to generalize across heterogeneous production environments. Fig. 7 highlights that these barriers are interdependent: weak data quality undermines generalizability, limited interpretability reduces operational trust, and computational constraints can prevent otherwise accurate models from being used in real-time field settings. Addressing these barriers is crucial for transitioning from experimental deployments to reliable, field-scale integration.

4.1. *Interpretability and accuracy*

One of the most prominent concerns surrounding AI/ML applications in engineering is interpretability. Traditional physics-based models, such as OLGA and PIPESIM, provide predictions grounded in well-understood conservation laws and empirical correlations. On the other hand, many machine learning models, especially deep learning architectures, operate as opaque systems, providing minimal insight into how they derive their outputs from their inputs. Although these models often achieve high predictive accuracy under conditions similar to their training data, their reliability can deteriorate when exposed to unrepresented or rare operational scenarios, such as sudden equipment failures or severe slugging events. This uncertainty contributes to a trust deficit, as engineers are understandably hesitant to depend on models that lack physical transparency, especially in safety-critical contexts where decisions directly influence equipment integrity, environmental protection, and production continuity. For this reason, uncertainty quantification must be treated as a core deployment requirement rather than an optional refinement. In practical surface-network applications, models should not only provide point predictions but also communicate confidence bounds, prediction intervals, or probability estimates that distinguish routine operating conditions from poorly represented or out-of-distribution scenarios. This is particularly important when AI outputs may influence choke adjustments, routing decisions, separator loading, or anomaly-response actions, where overconfident but incorrect recommendations can create both production losses and integrity risks. In such settings, uncertainty-aware models can support risk-based screening, escalation to human operators, and conservative fallback to physics-based or rule-based control when prediction confidence is low. To address this issue, emerging approaches such as PINNs, hybrid AI–physics frameworks, and explainable AI (XAI) techniques are being developed to enhance transparency and ensure physical consistency. However, despite their promise, the widespread industrial adoption of these interpretability-focused solutions remains limited. In this sense, trustworthy AI in surface-network operations is not expected to replace engineering judgment, but to function as a constrained and explainable decision-support layer whose recommendations remain auditable, physically defensible, and subject to operator oversight.

4.2. *Data quality and availability*

Despite significant advances in instrumentation and digitalization, data quality remains one of the most critical limitations for the deployment of AI and ML in oil and gas surface networks. The performance, stability, and generalizability of predictive models are highly dependent on the reliability, completeness, and representativeness of the underlying datasets, yet several persistent issues undermine data integrity:

- Noise and Sensor Drift: Field sensors frequently experience signal noise, calibration errors, and long-term drift. Variations in pressure, temperature, or flow measurements can introduce instability into AI predictions if not properly filtered or corrected. For example, Werneck et al. [62] explicitly applied anomaly removal and data augmentation to improve the robustness of short-term forecasting models trained on

producer and injector data, highlighting the practical importance of data cleaning in field-oriented applications.

- Missing and Incomplete Data: Sensor outages, communication dropouts, and transmission delays often create data gaps, particularly during critical transient operations such as pigging, startup, or shutdown. Conventional interpolation techniques may fail to capture the nonlinear dynamics of these events [91].
- Heterogeneous Sampling Rates: Different instruments operate at varying frequencies, complicating data synchronization and alignment for time-series modeling and reducing the temporal coherence of training datasets.
- Label Scarcity: Supervised learning models rely on labeled data, yet annotations for high-impact but infrequent events, such as hydrate formation, severe slugging, or sudden blockages, are scarce, limiting model accuracy and event detection capability. This challenge is evident in production-assurance studies such as Santos et al. [41], which used the Petrobras 3W benchmark [42] for classifying undesirable operating conditions, illustrating both the value and rarity of well-labeled operational event datasets.
- Bias and Representativeness: Historical datasets are typically dominated by normal operating conditions, underrepresenting anomalies and extreme transients. This imbalance can bias model training and reduce predictive performance under nonstandard conditions. Song et al. [59], for example, reported that some models were sensitive to high-productivity wells and showed overfitting tendencies, underscoring how nonrepresentative datasets can distort apparent model performance.
- Data Fragmentation: Operational data are often fragmented across departments, contractors, or proprietary systems, impeding the creation of unified, field-wide models and restricting access to diverse training datasets.

To mitigate these challenges, robust data preprocessing pipelines, including advanced noise filtering, gap-filling and imputation algorithms, data resampling, and anomaly detection, are essential. Moreover, semi-supervised, unsupervised, and transfer learning techniques can be explored to reduce dependence on labeled data and improve the adaptability of AI/ML models in dynamic surface-network environments. In newly developed fields or sensor-limited environments, the most stable options are often data-efficient classical ML models, hybrid gray-box frameworks, and physics-informed approaches, since these methods can leverage either structured low-volume datasets or prior physical knowledge more effectively than purely data-hungry deep-learning models.

4.3. Computational resources

AI/ML models, especially deep learning architectures, are often computationally intensive, posing significant challenges for large-scale, real-time deployment in oil and gas surface network operations. Training time is an important constraint, as high-fidelity models such as transformers, LSTMs, or DeepONets require extensive datasets and prolonged optimization cycles, which can limit their practicality in fast-paced operational settings. Resource limitations further compound the issue, since many remote production sites lack access to high-performance computing (HPC) infrastructure or reliable cloud connectivity. Moreover, real-time integration poses challenges, as surface network optimization frequently necessitates swift decision-making. However, computing latency in both training and inference can diminish responsiveness. In addition to technological limitations, energy use and costs are increasingly significant considerations, as training large AI models requires considerable processing power, thereby exacerbating sustainability and carbon footprint challenges. It is crucial to balance model complexity with feasibility to provide scalable, efficient, and environmentally sustainable AI/ML implementation in surface production systems.

4.4. Cybersecurity

Cybersecurity and operational integrity also represent important deployment barriers for AI/ML-enabled surface networks. As models become integrated with SCADA systems, digital twins, edge devices, and cloud-connected control architectures, the attack surface expands to include data tampering, malicious sensor spoofing, unauthorized model updates, and compromised control recommendations. These risks are particularly serious in closed-loop settings, where corrupted inputs or unsafe actuation logic may affect choke settings, compressor operation, routing decisions, or other safety-critical actions. Future deployment strategies must therefore incorporate secure data pipelines, access control, model-governance procedures, and fail-safe supervisory mechanisms to ensure that AI-assisted operations remain both reliable and defensible.

4.5. Generalizability

A significant challenge to scaling AI/ML across the oil and gas industry is the generalization problem, which refers to the difficulty of applying models trained in one field or asset to another with different geological, operational, or equipment characteristics. Most current AI/ML implementations are field-specific, tailored to individual assets and thus limited in their reusability across diverse production environments. This limitation is also evident in the reviewed literature. For example, Li et al. [66] trained a Transformer-based predictor on production data from only nine wells and validated it on six additional wells, while also relying only on production-rate time series, which limits broader transferability. Similarly, Huang et al. [67] reported strong forecasting performance on the Volve field using data from only three production wells, and Martínez and Rocha [61] showed that although their Seq2Seq GRU/LSTM framework performed extremely well on synthetic benchmark data, its accuracy declined on real-field Volve data because of greater operational complexity and shut-in behavior. As operating conditions evolve (through field maturation, rising water cuts, shifting gas–oil ratios, or the integration of new wells), flow dynamics change, and static models quickly become outdated unless they are continuously retrained. Moreover, unseen scenarios, such as hydrate blockages during extreme cold events or unexpected sand influx, are rarely represented in training datasets, further reducing model robustness. To partially address these challenges, transfer learning approaches that leverage pre-trained models followed by field-specific fine-tuning can be considered, allowing models to retain broadly learned production patterns while adapting to local conditions with limited additional data. To further improve robustness, hybrid approaches that combine data-driven learning with physics-based modeling have emerged as a promising solution, improving generalization by anchoring predictions in physical laws while adapting to field-specific data patterns. Overcoming this generalization barrier remains a crucial research priority for developing scalable, cross-asset AI/ML frameworks that support robust, real-time surface network optimization.

5. Future directions

The effective adoption of AI/ML for surface network optimization remains nascent, characterized by numerous pilot projects. However, to transition from isolated successes toward mainstream adoption, the industry must employ creative strategies that improve scalability, reliability, and interpretability. Fig. 8 summarizes the main strategic directions for advancing the field, emphasizing that future progress will

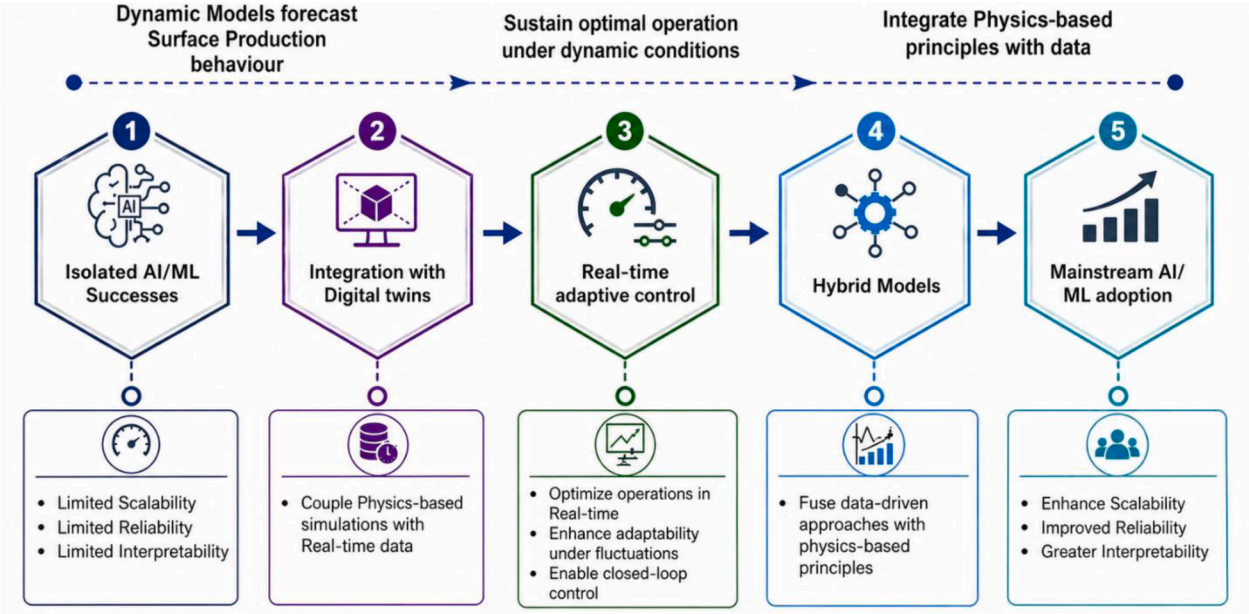


Fig. 8. Key future directions for advancing AI/ML applications in surface network optimization.

depend on integrating AI/ML with digital twins, adaptive control, physics-based constraints, and deployment-ready operational workflows.

5.1. Integration with digital twins

Digital twins, which are virtual representations of physical systems that continuously evolve based on real-time data, are becoming fundamental to next-generation oil and gas operations. When combined with AI/ML, digital twins of surface networks evolve into dynamic, living models that not only replicate present field conditions but also predict future behavior. Within this framework, AI/ML models serve as fast and adaptive surrogates, enabling rapid scenario testing, risk assessment, and optimization far beyond the speed and scalability of traditional simulators. The predictive capabilities of these AI-enhanced twins extend to anticipating flow instabilities, hydrate formation, or equipment failures, allowing operators to take proactive measures before disruptions occur. Moreover, through lifecycle adaptation, digital twins can remain continuously calibrated as networks evolve (whether through new tiebacks, water breakthrough, or equipment upgrades) by incorporating automated AI-driven retraining and data assimilation. Recent cross-domain reviews of digital twins in energy systems likewise emphasize that practical deployment depends on layered integration of sensing, data infrastructure, AI models, and cloud/edge computing across the asset lifecycle, while also facing recurring challenges related to model fidelity, interoperability, standardization, and cybersecurity [92].

The integration of AI/ML with physics-based modeling enables an effective, closed-loop system for real-time monitoring, diagnostics, and decision-making, hence improving operational efficiency, dependability, and long-term asset performance. In Fig. 8, the digital twins block is highlighted to show that it is the key step where AI/ML moves from isolated use cases to an integrated system that connects real-time data, physical models, and operational decision-making.

5.2. Real-time adaptive control and closed-loop optimization

Traditional optimization workflows in oil and gas operations often rely on batch analysis or periodic updates, limiting their ability to respond to rapidly changing field conditions. The future of surface-network management lies in real-time adaptive control, where AI and ML models interact directly with control systems to make continuous, data-driven adjustments across the production lifecycle. However, such closed-loop operation must be both risk-aware and uncertainty-aware. In practice, this means that AI-generated control actions should be filtered through operational constraints, confidence thresholds, actuator limits, and supervisory logic before implementation. Where uncertainty is high, the system should either recommend a conservative action, defer to a validated physics-based controller, or require operator confirmation rather than act autonomously. This type of safeguarded decision-making is especially important in multiphase surface networks, where local actions can propagate across wells, manifolds, and facilities and may trigger unintended pressure, flow-assurance, or integrity consequences.

Key enabling elements of this transition include the following:

- Dynamic Flow Control: Reinforcement learning and adaptive optimization algorithms can automatically adjust choke settings, compressor loads, and pigging schedules in real time to stabilize multiphase flow, prevent slugging, and maximize throughput.
- Anomaly Detection and Automated Response: Real-time ML models integrated into SCADA or distributed control systems can identify abnormal events, such as pressure surges, equipment malfunctions, or severe slugging, and trigger automated corrective actions before human intervention is required.
- Edge Computing Integration: Deploying lightweight ML models at the network edge, close to sensors and actuators, minimizes latency and ensures fast, localized control decisions, even in remote offshore environments with limited connectivity.
- Closed-Loop Frameworks: In a closed-loop configuration, AI/ML models not only analyze real-time sensor data but also generate optimization or control actions that are immediately implemented and validated through new feedback, allowing models to continuously learn from their interventions.
- Model Drift and Retraining: As operating conditions evolve, through reservoir pressure decline, water breakthrough, or network reconfiguration, AI models may experience concept drift, reducing their predictive accuracy. Continuous retraining with updated field data is essential to preserve model reliability and adaptability.
- Continuous Integration and Continuous Deployment (CI/CD): Incorporating CI/CD practices into ML pipelines enhances

reproducibility, version tracking, and automated retraining, ensuring that models remain aligned with current operational realities and resilient to data drift [93].

By moving beyond static prediction to active, adaptive, and self-improving control, AI/ML-driven closed-loop frameworks promise safer, more efficient, and autonomous operations, transforming real-time production optimization in surface networks.

5.3. Hybrid models

Although data-driven models offer significant computational advantages, their lack of physical grounding remains a concern for engineers, particularly in safety-critical, highly dynamic oil and gas operations. Consequently, future developments in surface network optimization are expected to focus on hybrid models that integrate physics-based principles with data-driven learning to achieve both accuracy and interpretability. PINNs represent one class of such hybrid models, where governing equations of fluid flow are incorporated into the learning process as residual constraints, ensuring that predictions remain consistent with fundamental conservation laws and reducing physically implausible outcomes. More general PIML frameworks extend this idea by embedding physical structure, states, or operators directly into the model formulation, rather than relying solely on residual-based constraints. Hybrid architecture also enhances uncertainty quantification by grounding predictions in established physical laws while accommodating field-specific anomalies observed in operational data. This makes hybrid models particularly attractive for risk-sensitive decision support, where both predictive performance and defensible confidence estimates are needed before recommendations can be trusted in operational workflows. Furthermore, their scalability makes them more transferable across assets, since the physics-based components remain consistent while data-driven modules adjust to local conditions. The future of surface network modeling and optimization will likely be influenced by hybrid AI-physics frameworks that effectively balance interpretability, predictive accuracy, and adaptability in intricate production platforms.

6. Conclusions

This review traces the evolution of surface-network modeling in the oil and gas industry, from empirical correlations and mechanistic simulators to steady-state and transient field solvers, ending with the rise of a new class of data-driven, hybrid, and physics-informed methods. The following are the main conclusions highlighting AI/ML opportunities in surface network optimization:

- AI/ML models can learn complex, nonlinear relationships in production systems that traditional physics-based simulators struggle to capture. They can infer unmeasured states, predict system behavior under uncertainty, and recommend optimal actions, thus enhancing and accelerating decision-making.
- Key applications include virtual flow metering, in which AI/ML enables accurate flow rate estimation using sensor data and historical records. In transient forecasting, sequence models (LSTMs, TCNs, Transformers) predict short-term changes in flow, pressure, and temperature, enabling proactive operations. Similarly, for nonlinear model predictive control, AI/ML models facilitate real-time optimization of choke settings, compressor loads, and routing decisions, improving throughput and reducing downtime.
- Advanced model types, such as ensemble methods, combine multiple models to improve prediction reliability and robustness. Sequence models such as LSTM, TCN, and Transformer architectures excel at capturing temporal dependencies in production data. GNNs model the topological relationships in surface networks, supporting network-level optimization and fault detection. PINNs and FNO/PINO integrate physical laws into ML models, ensuring predictions remain physically plausible and enabling fast, resolution-invariant simulations.
- For deployment and integration, AI/ML solutions are being architected for deployment across edge devices, on-premises servers, and cloud platforms, with CI/CD pipelines and drift monitoring for continuous improvement and reliability.
- Hybrid and physics-informed approaches combine mechanistic and data-driven components, leveraging the strengths of both to improve accuracy, interpretability, and generalizability. Physics-informed learning, such as PINNs, embeds governing equations directly into the training process, reducing the need for large labeled datasets and improving extrapolation to new scenarios.
- For real-time optimization and closed-loop control, AI/ML enables closed-loop systems that continuously analyze streaming data, detect anomalies, and autonomously adjust control actions to maintain optimal operation under dynamic conditions.
- Considering data-driven digital twins, integrating AI/ML enables dynamic, continuously updated virtual models of surface networks, supporting scenario analysis, risk assessment, and predictive maintenance.

To translate these advances into routine industrial practice, future work should prioritize four areas. First, the field needs larger field-validated datasets, shared benchmarking protocols, and more consistent reporting of evaluation metrics to enable credible cross-study comparison and stronger evidence of real-world performance. Second, more research is needed on uncertainty-aware and risk-aware decision-making so that predictive models can support safe closed-loop operation under rare events, changing operating conditions, and incomplete data. Third, hybrid physics–data-driven models should be advanced further to improve transferability across assets while preserving interpretability and physical consistency. Fourth, practical deployment research should focus on scalable integration across edge, on-premises, and cloud environments, supported by continuous monitoring, retraining, cybersecurity, and human oversight. Progress in these areas will be essential for moving from promising pilot applications to trustworthy, field-wide intelligent surface-network management. More broadly, the future of surface-network optimization will be shaped by the convergence of artificial intelligence, physics-based modeling, and advanced automation, enabling digital twins, real-time adaptive control, and increasingly predictive and autonomous production management. Overall, this review indicates that AI-driven, physics-grounded intelligence is not merely an enhancement to existing methods, but a key pathway toward smarter, safer, and more sustainable oil and gas production.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Ussama Ali: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paul Naveen:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Emad W. Al-Shalabi:** Writing – review &

editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Hamid Ait Abderrahmane:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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